WorkForce Health SA

Document Number: WFH-AFR-KUI-OD-26-001

Occupational Hygine

Operational Details

AFRIMAT

29 September 2026

AFRIMAT LTD
Afrimat CM North (Pty) Ltd
Kuipersbult Quarry
Remainder of the farm Kuipersbults 511LQ
LEPHALALE
Limpopo Province

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1 Legal structure

1A.1a	**[1]Legal entity registered name**	Afrimat CM North (Pty) Ltd

1A.1b	**Legal entity registered trade name**	Afrimat CM North (Pty) Ltd

1A.2	**Company registration number**	2002 / 025895 / 07

1A.3	**[2]Value added tax registration number**	4910205428

1A.4 Entity logo

1A.5a	**Registered physical address** (As per CIPC registration certificate)	3 Sennett Street **WORCESTER** Western Cape 0850

1A.5b	**Registered postal address** (As per CIPC registration certificate)	P O Box 505 **WORCESTER** Western Cape 6849

[1] CIPC registration certificate as Attachment A1.1
[2] SARS VAT registration certificate as Attachment A1.2

1A.5c	[3]**Physical address of the operation** (As per mining right / permit / license)	Remainder of the farm Kuipersbult 511 LQ, Lephalale, magisterial administrative district of Lephalale

1A.6	**Contact Official** (Employer Representative)
Initials and surname:	Mr C Reyneke
Telephone number:	+27 (0)12 664 5649
Facsimile:	None
Mobile number:	+27 (0)78 893 2725
E-mail:	Casper.reyneke@afrimat.co.za

1A.7	**Contact Official**
Initials and surname:	Mr Reinhard Stander
Telephone number:	+27 (0)76 233 6399
Facsimile:	None
Mobile number:	+27 (0)76 233 6399
E-mail:	Reinhard.stander@afrimat.co.za

1A.8	**Contact Official** (Accounts)
Initials and surname:	Mr Reinhard Stander
Telephone number:	+27 (0)76 233 6399
Facsimile:	None
Mobile number:	+27 (0)76 233 6399
E-mail:	Reinhard.stander@afrimat.co.za

[3] Mining right / permit / license as Attachment A1.3

1B.	**Mining authorisations / exemptions**	
1	**Mining permit number:**	No mining permit -refer to point 1B.2 below
2	**Mining right number:**	See Appendix A1.3
3	**Borehole registration number:**	None
4	**[4]Water usage license number:**	See Appendix A1.4
5	**Sunday work permission:**	None
6	**Exemptions** (specify)	None

2B.1	**Full registered name of control group**	Afrimat Ltd

2B.1a	**Legal entity registered name**	Afrimat Ltd

2B.1b	**Legal entity registered trade name**	Afrimat Ltd

2B.2	**[5]Company registration number**	2006 / 022534 / 06

2B.3	**[6]Value added tax registration number**	4200236356

2B.4 Control group logo

[4] Water usage license as Attachment A1.4
[5] Control group CIPC registration certificate as Attachment A1.5
[6] Control group SARS VAT registration certificate as Attachment A1.6

2B.5a	**Registered physical address** (As per CIPC registration certificate)	Tyger Valley Office Park Building number 2 Corner of Willie van Schoor and Old Oak Road **BELLVILLE** 7530

2B.5b	**Registered postal address** (As per CIPC registration certificate)	PO Box 5278 **TYGER VALLEY** 7536

2B.6	**Contact Official** (Chief Executive Officer)
Initials and surname:	Mr A J Van Heerden
Telephone number:	+27 (0)21 917 8840
Facsimile:	None
Mobile number:	Not available
E-mail:	andries@afrimat.co.za

2B.7	**Contact Official** (Accounts)
Initials and surname:	
Telephone number:	+27 (0)
Facsimile:	None
Mobile number:	+27 (0)
E-mail:	

2 Legal appointments

2A.1	**Owner** ***(Legal reference:*** *Section 2A(1))*
Initials and surname:	
Telephone number:	
Facsimile:	
Mobile number:	
E-mail:	
Letter of appointment submitted to the DMPR	

and / or

2A.2	**Chief excecutive officer** ***(Legal reference:*** *Section 2A(1))*
Initials and surname:	Mr A J Van Heerden
Telephone number:	+27 (0)21 917 8840
Facsimile:	None
Mobile number:	Not available
E-mail:	andries@afrimat.co.za
Letter of appointment submitted to the DMPR	Refer to letter of appointment

2A.3	**Employer representative** (***Legal reference:*** *Section 2A(2) or 2A(3) or 4(1))*
Initials and surname:	Mr C Reyneke
Telephone number:	+27 (0)12 664 5649
Facsimile:	None
Mobile number:	+27 (0)78 893 2725
E-mail:	Casper.reyneke@afrimat.co.za
Letter of appointment submitted to the DMPR	Refer to letter of appointment

2A.4	**Mine manager** (***Legal reference:*** *Section 3(1)*)
Initials and surname:	Mr P K Lekgeu
Telephone number:	+27 (0)60 742 8208
Facsimile:	None
Mobile number:	+27 (0)60 742 8208
E-mail:	Peter.lekgeu@afrimat.co.za
Letter of appointment submitted to the DMPR	Refer to letter of appointment

2A.5	**Occupational hygienist** (***Legal reference:*** *Section 12(1)*)
Initials and surname:	Mr N B Letlape
Telephone number:	+27 (0)67 418 4168
Facsimile:	None
Mobile number:	+27 (0)67 418 41689
E-mail:	ChiefOH@workforcehealthsa.co.za
Letter of appointment submitted to the DMPR	Refer to letter of appointment

2A.5.1	**Occupational hygienist assistant** (***Legal reference:*** *Section 9(2)*)
Initials and surname:	Mr N B Ngobese
Telephone number:	+27 (0)78 708 9415
Facsimile:	None
Mobile number:	+27 (0) 78 708 9415
E-mail:	admin@workforcehealthsa.co.za
Letter of appointment	WFH-AFR-KUI-LA-OH-ASS-26-001

2A.6	**Occupational medical practitioner** (***Legal reference:*** *Section 13(3)(a)(i)*)
Initials and surname:	Dr K O Sebopa
Telephone number:	+27 (0)82 925 1158
Facsimile:	None
Mobile number:	+27 (0) 82 925 1158
E-mail:	drmasenyaolivia@gmail.com
Letter of appointment submitted to the DMPR	Refer to letter of appointment

2A.6.1	**Occupational health practitioner** (***Legal reference:*** *Section 9(2)*)
Initials and surname:	
Telephone number:	
Facsimile:	
Mobile number:	
E-mail:	
Letter of appointment	

2A.7	**Engineer** (***Legal reference:*** *Schedule 4, Regulation 2.13.1)*
Initials and surname:	
Telephone number:	
Facsimile:	
Mobile number:	
E-mail:	
Letter of appointment submitted to the DMPR	

2A.7.1	**Competent person – Mechanical** (***Legal reference:*** *Schedule 4, Regulation 2.13.2)*
Initials and surname:	Mr H J Pienaar
Telephone number:	+27 (0)78 457 2443
Facsimile:	None
Mobile number:	+27 (0)78 457 2443
E-mail:	henniepienaarhh@gmail.com
Letter of appointment submitted to the DMPR	Refer to letter of appointment

2A.7.2	**Competent person – Electrical** (***Legal reference:*** *Schedule 4, Regulation 2.13.2)*
Initials and surname:	Mr J M A Spangenberg
Telephone number:	+27 (0)73 527 1840
Facsimile:	None
Mobile number:	+27 (0) 73 527 1840
E-mail:	ditshabaelectrical@gmail.com
Letter of appointment submitted to the DMPR	Refer to letter of appointment

2A.8.1	**Occupational health (hygiene and medicine) mandatory code of practice drafting / management / risk assessment committee member** *(**Legal reference:** Section 9(2) – Mine manager)*
Initials and surname:	Mr P K Lekgeu
Telephone number:	+27 (0)60 742 8208
Facsimile:	None
Mobile number:	+27 (0)60 742 8208
E-mail:	Peter.lekgeu@afrimat.co.za
Letter of appointment	WFH-AFR-KUI-LA-OHS-DCM-26-001.1

2A.8.2	**Occupational health (hygiene and medicine) mandatory code of practice drafting / management / risk assessment committee member** *(**Legal reference:** Section 9(2) – Occupational Hygienist)*
Initials and surname:	Mr N B Letlape
Telephone number:	+27 (0)67 418 4168
Facsimile:	None
Mobile number:	+27 (0)67 418 41689
E-mail:	ChiefOH@workforcehealthsa.co.za
Letter of appointment	

2A.8.3	**Occupational health (hygiene and medicine) mandatory code of practice drafting / management / risk assessment committee member** *(**Legal reference:** Section 9(2) – Occupational Medical Practitioner)*
Initials and surname:	Dr K O Sebopa
Telephone number:	+27 (0)82 925 1158
Facsimile:	None
Mobile number:	+27 (0) 82 925 1158
E-mail:	drmasenyaolivia@gmail.com
Letter of appointment	

2A.8.4	**Occupational health (hygiene and medicine) mandatory code of practice drafting / management / risk assessment committee member** *(**Legal reference:** Section 9(2) – Occupational Hygienist Assistant)*
Initials and surname:	Mr N B Ngobese
Telephone number:	+27 (0)78 708 9415
Facsimile:	None
Mobile number:	+27 (0) 78 708 9415
E-mail:	admin@workforcehealthsa.co.za
Letter of appointment	

2A.8.5	**Occupational health (hygiene and medicine) mandatory code of practice drafting / management / risk assessment committee member** *(**Legal reference:** Section 9(2) – Competent Person - Mechanical)*
Initials and surname:	Mr H J Pienaar
Telephone number:	+27 (0)78 457 2443
Facsimile:	None
Mobile number:	+27 (0)78 457 2443
E-mail:	henniepienaarhh@gmail.com
Letter of appointment	

2A.8.6	**Occupational health (hygiene and medicine) mandatory code of practice drafting / management / risk assessment committee member** *(**Legal reference:** Section 9(2) – Competent Person - Electrical)*
Initials and surname:	Mr J M A Spangenberg
Telephone number:	+27 (0)73 527 1840
Facsimile:	None
Mobile number:	+27 (0) 73 527 1840
E-mail:	ditshabaelectrical@gmail.com
Letter of appointment	

2A.8.7	**Occupational health (hygiene and medicine) mandatory code of practice drafting / management / risk assessment committee member** *(**Legal reference:** Section 9(2) – Safety Officer (Intern))*
Initials and surname:	Ms W O Mandoma
Telephone number:	+27 (0)78 226 6304
Facsimile:	None
Mobile number:	+27 (0) 78 226 6304
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of appointment	

2A.8.8	**Occupational health (hygiene and medicine) mandatory code of practice drafting / management / risk assessment committee member** *(**Legal reference:** Section 9(2) – Health and Safety Representative)*
Initials and surname:	Ms W G Jordaan
Telephone number:	+27 (0)76 224 9818
Facsimile:	None
Mobile number:	+27 (0) 76 224 9818
E-mail:	lephalaleRMC2@afrimat.co.za
Letter of appointment	

2A.9	**Health and safety representative** (**Legal reference:** *(Section 29(4), Regulation 6.9)*
Initials and surname:	Ms W G Jordaan
Telephone number:	+27 (0)76 224 9818
Facsimile:	None
Mobile number:	+27 (0) 76 224 9818
E-mail:	lephalaleRMC2@afrimat.co.za
Letter of appointment	

2A.10.1	**Respiratory conservation committee member** *(**Legal reference:** Section 9(2), MCOP airborne pollutants – Mine Manager)*
Initials and surname:	Mr P K Lekgeu
Telephone number:	+27 (0)60 742 8208
Facsimile:	None
Mobile number:	+27 (0)60 742 8208
E-mail:	Peter.lekgeu@afrimat.co.za
Letter of appointment	

2A.10.2	**Respiratory conservation committee member** *(**Legal reference:** Section 9(2), MCOP airborne pollutants – Occupational Hygienist)*
Initials and surname:	Mr N B Letlape
Telephone number:	+27 (0)67 418 4168
Facsimile:	None
Mobile number:	+27 (0)67 418 41689
E-mail:	ChiefOH@workforcehealthsa.co.za
Letter of appointment	

2A.10.3	**Respiratory conservation committee member** *(**Legal reference:** Section 9(2) , MCOP airborne pollutants – Occupational Hygienist Assistant)*
Initials and surname:	Mr N B Ngobese
Telephone number:	+27 (0)78 708 9415
Facsimile:	None
Mobile number:	+27 (0) 78 708 9415
E-mail:	admin@workforcehealthsa.co.za
Letter of appointment	

2A.10.4	**Respiratory conservation committee member** *(**Legal reference:** Section 9(2) , MCOP airborne pollutants – Occupational Medical Practitioner)*
Initials and surname:	Dr K O Sebopa
Telephone number:	+27 (0)82 925 1158
Facsimile:	None
Mobile number:	+27 (0) 82 925 1158
E-mail:	drmasenyaolivia@gmail.com
Letter of appointment	

2A.10.5	**Respiratory conservation committee member** *(**Legal reference:** Section 9(2), MCOP airborne pollutants – Safety Officer (intern))*
Initials and surname:	Ms W O Mandoma
Telephone number:	+27 (0)78 226 6304
Facsimile:	None
Mobile number:	+27 (0) 78 226 6304
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of appointment	

2A.10.6	**Respiratory conservation committee member** *(**Legal reference:** Section 9(2), MCOP airborne pollutants – Competent Person - Mechanical)*
Initials and surname:	Mr H J Pienaar
Telephone number:	+27 (0)78 457 2443
Facsimile:	None
Mobile number:	+27 (0)78 457 2443
E-mail:	henniepienaarhh@gmail.com
Letter of appointment	

2A.10.7	**Respiratory conservation committee member** *(**Legal reference:** Section 9(2), MCOP airborne pollutants – Competent Person - Electrical)*
Initials and surname:	Mr J M A Spangenberg
Telephone number:	+27 (0)73 527 1840
Facsimile:	None
Mobile number:	+27 (0) 73 527 1840
E-mail:	ditshabaelectrical@gmail.com
Letter of appointment	

2A.10.8	**Respiratory conservation committee member** *(**Legal reference:** Section 9(2), MCOP airborne pollutants – Safety Representative)*
Initials and surname:	Ms W G Jordaan
Telephone number:	+27 (0)76 224 9818
Facsimile:	None
Mobile number:	+27 (0) 76 224 9818
E-mail:	lephalaleRMC2@afrimat.co.za
Letter of appointment	

2A.11.1	**Hearing conservation committee member** *(**Legal reference:** Section 9(2), MCOP noise – Mine Manager)*
Initials and surname:	Mr P K Lekgeu
Telephone number:	+27 (0)60 742 8208
Facsimile:	None
Mobile number:	+27 (0)60 742 8208
E-mail:	Peter.lekgeu@afrimat.co.za
Letter of appointment	

2A.11.2	**Hearing conservation committee member** *(**Legal reference:** Section 9(2), MCOP noise – Occupational Hygienist)*
Initials and surname:	Mr N B Letlape
Telephone number:	+27 (0)67 418 4168
Facsimile:	None
Mobile number:	+27 (0)67 418 41689
E-mail:	ChiefOH@workforcehealthsa.co.za
Letter of appointment	

2A.11.3	**Hearing conservation committee member** *(**Legal reference:** Section 9(2) , MCOP noise – Occupational Hygienist Assistant)*
Initials and surname:	Mr N B Ngobese
Telephone number:	+27 (0)78 708 9415
Facsimile:	None
Mobile number:	+27 (0) 78 708 9415
E-mail:	admin@workforcehealthsa.co.za
Letter of appointment	

2A.11.4	**Hearing conservation committee member** *(**Legal reference:** Section 9(2) , MCOP noise – Occupational Medical Practitioner)*
Initials and surname:	Dr K O Sebopa
Telephone number:	+27 (0)82 925 1158
Facsimile:	None
Mobile number:	+27 (0) 82 925 1158
E-mail:	drmasenyaolivia@gmail.com
Letter of appointment	

2A.11.5	**Hearing conservation committee member** *(**Legal reference:** Section 9(2) , MCOP noise – Safety Officer (intern))*
Initials and surname:	Ms W O Mandoma
Telephone number:	+27 (0)78 226 6304
Facsimile:	None
Mobile number:	+27 (0) 78 226 6304
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of appointment	

2A.11.6	**Hearing conservation committee member** *(**Legal reference:** Section 9(2), MCOP noise – Competent Person - Mechanical)*
Initials and surname:	Mr H J Pienaar
Telephone number:	+27 (0)78 457 2443
Facsimile:	None
Mobile number:	+27 (0)78 457 2443
E-mail:	henniepienaarhh@gmail.com
Letter of appointment	

2A.11.7	**Hearing conservation committee member** *(**Legal reference:** Section 9(2), MCOP noise – Competent Person - Electrical)*
Initials and surname:	Mr J M A Spangenberg
Telephone number:	+27 (0)73 527 1840
Facsimile:	None
Mobile number:	+27 (0) 73 527 1840
E-mail:	ditshabaelectrical@gmail.com
Letter of appointment	

2A.11.8	**Hearing conservation committee member** *(**Legal reference:** Section 9(2), MCOP noise – Safety Representative)*
Initials and surname:	Ms W G Jordaan
Telephone number:	+27 (0)76 224 9818
Facsimile:	None
Mobile number:	+27 (0) 76 224 9818
E-mail:	lephalaleRMC2@afrimat.co.za
Letter of appointment	

2A.12.1	**Fatigue management committee member** *(**Legal reference:** Section 9(2), MCOP fatigue management – Mine Manager)*
Initials and surname:	Mr P K Lekgeu
Telephone number:	+27 (0)60 742 8208
Facsimile:	None
Mobile number:	+27 (0)60 742 8208
E-mail:	Peter.lekgeu@afrimat.co.za
Letter of appointment	

2A.12.2	**Fatigue management committee member** *(**Legal reference:** Section 9(2), MCOP fatigue management – Occupational Hygienist)*
Initials and surname:	Mr N B Letlape
Telephone number:	+27 (0)67 418 4168
Facsimile:	None
Mobile number:	+27 (0)67 418 41689
E-mail:	ChiefOH@workforcehealthsa.co.za
Letter of appointment	

2A.12.3	**Fatigue management committee member** *(**Legal reference:** Section 9(2) , MCOP fatigue management – Occupational Hygienist Assistant)*
Initials and surname:	Mr N B Ngobese
Telephone number:	+27 (0)78 708 9415
Facsimile:	None
Mobile number:	+27 (0) 78 708 9415
E-mail:	admin@workforcehealthsa.co.za
Letter of appointment	

2A.12.4	**Fatigue management committee member** *(**Legal reference:** Section 9(2) , MCOP fatigue management – Occupational Medical Practitioner)*
Initials and surname:	Dr K O Sebopa
Telephone number:	+27 (0)82 925 1158
Facsimile:	None
Mobile number:	+27 (0) 82 925 1158
E-mail:	drmasenyaolivia@gmail.com
Letter of appointment	

2A.12.5	**Fatigue management committee member** *(**Legal reference:** Section 9(2) , MCOP fatigue management – Safety Officer (intern)*
Initials and surname:	Ms W O Mandoma
Telephone number:	+27 (0)78 226 6304
Facsimile:	None
Mobile number:	+27 (0) 78 226 6304
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of appointment	

2A.12.6	**Fatigue management committee member** *(**Legal reference:** Section 9(2), MCOP fatigue management – Competent Person - Mechanical)*
Initials and surname:	Mr H J Pienaar
Telephone number:	+27 (0)78 457 2443
Facsimile:	None
Mobile number:	+27 (0)78 457 2443
E-mail:	henniepienaarhh@gmail.com
Letter of appointment	

2A.12.7	**Fatigue management committee member** *(**Legal reference:** Section 9(2), MCOP fatigue management – Competent Person - Electrical)*
Initials and surname:	Mr J M A Spangenberg
Telephone number:	+27 (0)73 527 1840
Facsimile:	None
Mobile number:	+27 (0) 73 527 1840
E-mail:	ditshabaelectrical@gmail.com
Letter of appointment	

2A.12.8	**Fatigue management committee member** *(**Legal reference:** Section 9(2), MCOP fatigue management – Safety Representative)*
Initials and surname:	Ms W G Jordaan
Telephone number:	+27 (0)76 224 9818
Facsimile:	None
Mobile number:	+27 (0) 76 224 9818
E-mail:	lephalaleRMC2@afrimat.co.za
Letter of appointment	

2A.13.1	**Thermal stress management committee member** *(**Legal reference:** Section 9(2), MCOP fatigue management – Mine Manager)*
Initials and surname:	Mr P K Lekgeu
Telephone number:	+27 (0)60 742 8208
Facsimile:	None
Mobile number:	+27 (0)60 742 8208
E-mail:	Peter.lekgeu@afrimat.co.za
Letter of appointment	

2A.13.2	**Thermal stress management committee member** *(**Legal reference:** Section 9(2), MCOP fatigue management – Occupational Hygienist)*
Initials and surname:	Mr N B Letlape
Telephone number:	+27 (0)67 418 4168
Facsimile:	None
Mobile number:	+27 (0)67 418 41689
E-mail:	ChiefOH@workforcehealthsa.co.za
Letter of appointment	

2A.13.3	**Thermal stress management committee member** *(**Legal reference:** Section 9(2) , MCOP fatigue management – Occupational Hygienist Assistant)*
Initials and surname:	Mr N B Ngobese
Telephone number:	+27 (0)78 708 9415
Facsimile:	None
Mobile number:	+27 (0) 78 708 9415
E-mail:	admin@workforcehealthsa.co.za
Letter of appointment	

2A.13.4	**Thermal stress management committee member** *(**Legal reference:** Section 9(2) , MCOP fatigue management – Occupational Medical Practitioner)*
Initials and surname:	Dr K O Sebopa
Telephone number:	+27 (0)82 925 1158
Facsimile:	None
Mobile number:	+27 (0) 82 925 1158
E-mail:	drmasenyaolivia@gmail.com
Letter of appointment	

2A.13.5	**Thermal stress management committee member** *(**Legal reference:** Section 9(2) , MCOP fatigue management – Safety Officer (intern)*
Initials and surname:	Ms W O Mandoma
Telephone number:	+27 (0)78 226 6304
Facsimile:	None
Mobile number:	+27 (0) 78 226 6304
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of appointment	

2A.13.6	**Thermal stress management committee member** *(**Legal reference:** Section 9(2), MCOP fatigue management – Competent Person - Mechanical)*
Initials and surname:	Mr H J Pienaar
Telephone number:	+27 (0)78 457 2443
Facsimile:	None
Mobile number:	+27 (0)78 457 2443
E-mail:	henniepienaarhh@gmail.com
Letter of appointment	

2A.13.7	**Thermal stress management committee member** *(**Legal reference:** Section 9(2), MCOP fatigue management – Competent Person - Electrical)*
Initials and surname:	Mr J M A Spangenberg
Telephone number:	+27 (0)73 527 1840
Facsimile:	None
Mobile number:	+27 (0) 73 527 1840
E-mail:	ditshabaelectrical@gmail.com
Letter of appointment	

2A.13.8	**Thermal stress management committee member** *(**Legal reference:** Section 9(2), MCOP fatigue management – Safety Representative)*
Initials and surname:	Ms W G Jordaan
Telephone number:	+27 (0)76 224 9818
Facsimile:	None
Mobile number:	+27 (0) 76 224 9818
E-mail:	lephalaleRMC2@afrimat.co.za
Letter of appointment	

2A.14	**Mine surveyor** (***Legal reference:** Regulation 17)*
Initials and surname:	Mr G Brent-meek
Telephone number:	+27 (0)82 776 3607
Facsimile:	None
Mobile number:	+27 (0)82 776 3607
E-mail:	gbm@belanet.co.za
Letter of appointment submitted to the DMPR	

2A.15	**Rock engineer** (***Legal reference:** Regulation 14.1, 14.1(6), 14.1(8))*
Initials and surname:	Mr M Steinman
Telephone number:	+27 (0)82 378 9378
Facsimile:	None
Mobile number:	+27 (0)82 378 9378
E-mail:	steinmanngsp@yahoo.com
Letter of appointment submitted to the DMPR	

2A.16	**Occupational hygiene system administrator** (***Legal reference:*** *Section 9(2), MCOP quality assurance)*
Initials and surname:	Ms W O Mandoma
Telephone number:	+27 (0)78 226 6304
Facsimile:	None
Mobile number:	+27 (0) 78 226 6304
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of appointment	

2A.17	**Emergency preparedness and response coordinator** ***Legal reference:*** *Section 9(2), MCOP emergency preparedness and response)*
Initials and surname:	Ms W G Jordaan
Telephone number:	+27 (0)76 224 9818
Facsimile:	None
Mobile number:	+27 (0) 76 224 9818
E-mail:	lephalaleRMC2@afrimat.co.za
Letter of appointment	

2A.18.1	**First aider** ***Legal reference:*** *Section 9(2) – Emergency preparedness and response MCOP)*
Initials and surname:	Ms W G Jordaan
Telephone number:	+27 (0)76 224 9818
Facsimile:	None
Mobile number:	+27 (0) 76 224 9818
E-mail:	lephalaleRMC2@afrimat.co.za
Letter of appointment	

2A.18.2	**First aider** *(**Legal reference:** Section 9(2) – Emergency preparedness and response MCOP)*
Initials and surname:	Mr M S Kekae
Telephone number:	+27 (0)78 226 6304
Facsimile:	None
Mobile number:	+27 (0) 78 226 6304
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of appointment	

2A.18.3	**First aider** *(**Legal reference:** Section 9(2) – Emergency preparedness and response MCOP)*
Initials and surname:	Mr A N Seodisa
Telephone number:	+27 (0)71 623 4161
Facsimile:	None
Mobile number:	+27 (0) 71 623 4161
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of appointment	

2A.19.1	**Fire fighter** *(**Legal reference:** Section 9(2) – Emergency preparedness and response MCOP and prevention of fires at mines)*
Initials and surname:	Ms W G Jordaan
Telephone number:	+27 (0)76 224 9818
Facsimile:	None
Mobile number:	+27 (0)76 224 9818
E-mail:	lephalaleRMC2@afrimat.co.za
Letter of appointment	

2A.19.2	**Fire fighter** *(**Legal reference:** Section 9(2) – Emergency preparedness and response MCOP and prevention of fires at mines)*
Initials and surname:	Mr O D Molekoa
Telephone number:	+27 (0)67 996 1357
Facsimile:	None
Mobile number:	+27 (0)67 996 1357
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of appointment	

2A.19.3	**Fire fighter** *(**Legal reference:** Section 9(2) – Emergency preparedness and response MCOP and prevention of fires at mines)*
Initials and surname:	Mr G Matlala
Telephone number:	+27 (0)76 996 1401
Facsimile:	None
Mobile number:	+27 (0)76 996 1401
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of appointment	

3 Other legal / operational appointments

3A.1	**Chief safety officer** (***Legal reference:*** *Schedule 4, Regulation 2.17.4)*
Initials and surname:	
Telephone number:	
Facsimile:	
Mobile number:	
E-mail:	
Letter of appointment	

3A.2	**Safety officer** (***Legal reference:*** *Schedule 4, Regulation 2.17.1)*
Initials and surname:	Ms W O Mandoma
Telephone number:	+27 (0)78 226 6304
Facsimile:	None
Mobile number:	+27 (0) 78 226 6304
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of appointment	

3A.3	**Sub-ordinate manager** (***Legal reference:*** *Schedule 4, Regulation 2.6.1)*
Initials and surname:	
Telephone number:	
Facsimile:	
Mobile number:	
E-mail:	
Letter of appointment	

3A.4	**Assistant to the manager (Team leader / Supervisor)** (***Legal reference:*** *Schedule 4, Regulation 2.9.2)*
Initials and surname:	Mr M K Mphogo
Telephone number:	+27 (0)82 605 5779
Facsimile:	None
Mobile number:	+27 (0)82 605 5779
E-mail:	lephalalermc@afrimat.co.za
Letter of appointment	

3A.5	**Blaster** (***Legal reference:*** *Regulation 4.4(1))*
Initials and surname:	
Telephone number:	
Facsimile:	
Mobile number:	
E-mail:	
Letter of appointment	

3A.6	**Team leader (blaster assistant)** (**Legal reference:** *Regulation 4.4(3) and Schedule 4, Regulation 2.9.2)*
Initials and surname:	
Telephone number:	
Facsimile:	
Mobile number:	
E-mail:	
Letter of appointment	

4 Legal authorisations

4A.1.1	**Trackless mobile equipment operator** (**Legal reference:** *Regulation 8.10 (23.4) and 8.10 (23.5) and Schedule 4, Minerals Regulation 18.1.7 and 18.2.1*)
Initials and surname:	Mr N H Nkuna (fron-end loader)
Telephone number:	+27 (0)73 371 4640
Facsimile:	None
Mobile number:	+27 (0) 73 371 4640
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of authorisation	

4A.1.2	**Trackless mobile equipment operator** (**Legal reference:** *Regulation 8.10 (23.4) and 8.10 (23.5) and Schedule 4, Minerals Regulation 18.1.7 and 18.2.1*)
Initials and surname:	Mr E T Mashasha (front-end loader)
Telephone number:	+27 (0)76 085 1873
Facsimile:	None
Mobile number:	+27 (0) 76 085 1873
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of authorisation	

4A.1.3	**Trackless mobile equipment operator** (**Legal reference:** *Regulation 8.10 (23.4) and 8.10 (23.5) and Schedule 4, Minerals Regulation 18.1.7 and 18.2.1*)
Initials and surname:	Mr O D Molekoa (back tipper truck)
Telephone number:	+27 (0)67 996 1357
Facsimile:	None
Mobile number:	+27 (0)67 996 1357
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of authorisation	

4A.1.4	**Trackless mobile equipment operator** (**Legal reference:** *Regulation 8.10 (23.4) and 8.10 (23.5) and Schedule 4, Minerals Regulation 18.1.7 and 18.2.1*)
Initials and surname:	Mr G Matlala (front-end loader)
Telephone number:	+27 (0)76 996 1401
Facsimile:	None
Mobile number:	+27 (0)76 996 1401
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of authorisation	

4A.2.1	**Authorisation to render ineffective, remove, and / or interfere with anything provided for health and safety** (**Legal reference:** *Schedule 4, Minerals Regulation 3.14 (b)(c)*)
Initials and surname:	Mr N H Nkuna (fron-end loader)
Telephone number:	+27 (0)73 371 4640
Facsimile:	None
Mobile number:	+27 (0) 73 371 4640
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of authorisation	

4A.2.2	**Authorisation to render ineffective, remove, and / or interfere with anything provided for health and safety** (**Legal reference:** *Schedule 4, Minerals Regulation 3.14 (b)(c)*)
Initials and surname:	Mr E T Mashasha (front-end loader)
Telephone number:	+27 (0)76 085 1873
Facsimile:	None
Mobile number:	+27 (0) 76 085 1873
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of authorisation	

4A.2.3	**Authorisation to render ineffective, remove, and / or interfere with anything provided for health and safety** (**Legal reference:** *Schedule 4, Minerals Regulation 3.14 (b)(c)*)
Initials and surname:	Mr O D Molekoa (back tipper truck)
Telephone number:	+27 (0)67 996 1357
Facsimile:	None
Mobile number:	+27 (0)67 996 1357
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of authorisation	

4A.2.4	**Authorisation to render ineffective, remove, and / or interfere with anything provided for health and safety** (**Legal reference:** *Schedule 4, Minerals Regulation 3.14 (b)(c)*)
Initials and surname:	Mr G Matlala (front-end loader)
Telephone number:	+27 (0)76 996 1401
Facsimile:	None
Mobile number:	+27 (0)76 996 1401
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of authorisation	

4A.3.1	**Authorisation to start, operate and maintain machinery** (**Legal reference:** *Regulation 8.8(3)(a)(c) and 8.9(1)(i)*)
Initials and surname:	Mr N H Nkuna (fron-end loader)
Telephone number:	+27 (0)73 371 4640
Facsimile:	None
Mobile number:	+27 (0) 73 371 4640
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of authorisation	

4A.3.2	**Authorisation to start, operate and maintain machinery** (**Legal reference:** *Regulation 8.8(3)(a)(c) and 8.9(1)(i)*)
Initials and surname:	Mr E T Mashasha (front-end loader)
Telephone number:	+27 (0)76 085 1873
Facsimile:	None
Mobile number:	+27 (0) 76 085 1873
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of authorisation	

4A.3.3	**Authorisation to start, operate and maintain machinery** (**Legal reference:** *Regulation 8.8(3)(a)(c) and 8.9(1)(i)*)
Initials and surname:	Mr O D Molekoa (back tipper truck)
Telephone number:	+27 (0)67 996 1357
Facsimile:	None
Mobile number:	+27 (0)67 996 1357
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of authorisation	

4A.3.4	**Authorisation to start, operate and maintain machinery** (**Legal reference:** *Regulation 8.8(3)(a)(c) and 8.9(1)(i)*)
Initials and surname:	Mr G Matlala (front-end loader)
Telephone number:	+27 (0)76 996 1401
Facsimile:	None
Mobile number:	+27 (0)76 996 1401
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of authorisation	

4A.3.5	**Authorisation to start, operate and maintain machinery** (**Legal reference:** *Regulation 8.8(3)(a)(c) and 8.9(1)(i)*)
Initials and surname:	Mr A N Seodisa
Telephone number:	+27 (0)71 623 4161
Facsimile:	None
Mobile number:	+27 (0) 71 623 4161
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of authorisation	

4A.3.6	**Authorisation to start, operate and maintain machinery** (**Legal reference:** *Regulation 8.8(3)(a)(c) and 8.9(1)(i)*)
Initials and surname:	Mr M S Kekae
Telephone number:	+27 (0)78 226 6304
Facsimile:	None
Mobile number:	+27 (0) 78 226 6304
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of authorisation	

4A.3.7	**Authorisation to start, operate and maintain machinery** (**Legal reference:** *Regulation 8.8(3)(a)(c) and 8.9(1)(i)*)
Initials and surname:	Mr K M Motshegwa
Telephone number:	+27 (0)76 068 2862
Facsimile:	None
Mobile number:	+27 (0) 76 068 2862
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of authorisation	

4A.4.1	**Authorisation to operate, maintain, clean, and repair conveyor belt installations** (**Legal reference:** *Regulation 8.9(1)(i)*)
Initials and surname:	Mr A N Seodisa
Telephone number:	+27 (0)71 623 4161
Facsimile:	None
Mobile number:	+27 (0) 71 623 4161
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of authorisation	

4A.4.1	**Authorisation to operate, maintain, clean, and repair conveyor belt installations** (**Legal reference:** *Regulation 8.9(1)(i)*)
Initials and surname:	Mr M S Kekae
Telephone number:	+27 (0)78 226 6304
Facsimile:	None
Mobile number:	+27 (0) 78 226 6304
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of authorisation	

4A.4.1	**Authorisation to operate, maintain, clean, and repair conveyor belt installations** (**Legal reference:** *Regulation 8.9(1)(i)*)
Initials and surname:	Mr K M Motshegwa
Telephone number:	+27 (0)76 068 2862
Facsimile:	None
Mobile number:	+27 (0) 76 068 2862
E-mail:	Winnie.mandoma@afrimat.co.za
Letter of authorisation	

5 Mining details

5.1	Classification of concern	Commodity code	DMPR mine code	DMPR region and office
	Mine			Limpopo region / Polokwane office

5.2 Operation description

Afrimat Ltd, Afrimat CM North (Pty) Ltd, Kuipersbult Quarry is an opencast quarzitic sandstone and conglomerate aggregate mine with a ready-mix plant. The process entails the following: -

- **Opencast mining**
  - Clearing of mining area;
  - Drilling; and
  - Blasting.

- **Operation of several self-propelled mobile machines**
  - Loading of mined ore in the pit using an excavator, onto haul trucks;
  - Hauling from the pit to stockpiles / plant area;
  - Feeding of crusher with an excavator / front-end loader;
  - Loading of final product, by means of front-end loader, onto haul trucks – final product stockpiles;
  - Hauling of final product to stockpiles – various product sizes;
  - Loading of final product, by means of front-end loader, onto client transport trucks;
  - Loading of aggregate, by means of front-end loader, and feeding ready-mix batching plant;
  - Watering of haul roads with a water bowser / water trailer and all areas requiring watering down to prevent dust generation from the use of mine / other vehicles; and
  - Ready-mix trucks are used for transportation of concrete to clients.

- **Crushing and screening plant**
  - Crushing – primary (jaw) and secondary (cone) crusher/s;
  - Screening of crushed ore; and
  - Stock piling of final product - exit conveyor/s.

- **Ready-mix batching plant**
  - Batching (mixing) various concrete mixtures; and
  - Ready-mix trucks for transport to clients.

- **Management and administration**
  - Management of operation;
  - Support services performed by: - geologist, rock engineer, surveyor. etc.; and
  - Conducting several administrative duties / tasks.

- **Support services (engineering maintenance)**
  - Several maintenance and engineering tasks.

The operation of Afrimat Ltd, Afrimat CM North (Pty) Ltd, Kuipersbult Quarry, is located on the remainder of the farm Kuipersbult in the magisterial district of Lephalale in the Limpopo province.

The mining rights are owned by Afrimat Aggregate Trading (Pty) Ltd. An approximate location, with the global positioning system coordinates (GPS) for the mine (-23.70760° S, 27.57939° E) and shown on the map below.

5.3	**Location of Afrimat Ltd, Afrimat CM North (Pty) Ltd, Kuipersbult Quarry**



5.4	**Commodoties produced:**	Various sized aggregate and ready-mix concrete

5.5	Mining and processing methods (as per process flow)

- **Opencast mining**
  - Clearing of mining area;
  - Drilling; and
  - Blasting.

- **Operation of several self-propelled mobile machines**
  - Loading of mined ore in the pit using an excavator, onto haul trucks;
  - Hauling from the pit to stockpiles / plant area;
  - Feeding of crusher with an excavator / front-end loader;
  - Loading of final product, by means of front-end loader, onto haul trucks – final product stockpiles;
  - Hauling of final product to stockpiles – various product sizes;
  - Loading of final product, by means of front-end loader, onto client transport trucks;
  - Loading of aggregate, by means of front-end loader, and feeding ready-mix batching plant;
  - Watering of haul roads with a water bowser / water trailer and all areas requiring watering down to prevent dust generation from the use of mine / other vehicles; and
  - Ready-mix trucks are used for transportation of concrete to clients.

- **Crushing and screening plant**
  - Crushing – primary (jaw) and secondary (cone) crusher/s;
  - Screening of crushed ore; and
  - Stock piling of final product - exit conveyor/s.

- **Ready-mix batching plant**
  - Batching (mixing) various concrete mixtures; and
  - Ready-mix trucks for transport to clients.

- **Management and administration**
  - Management of operation;
  - Support services performed by: - geologist, rock engineer, surveyor. etc.; and
  - Conducting several administrative duties / tasks.

- **Support services (engineering maintenance)**
  - Several maintenance and engineering tasks.

6a Areas, activity codes, and homogeneous exposure groups

Area number and description:	1 – Kuipersbult Quarry - Opencast
Homogeneous exposure group number	**Description**
1	Drilling
2	Blasting
Activity code (as per DME codebook)	**Description**
30	Roving plant
07	Opencast

Area number and description:	2 – Kuipersbult Quarry – self-propelled mobile machines
Homogeneous exposure group number	**Description**
1	Self-propelled mobile machines, excluding drill rig/s (roving plant)
Activity code (as per DME codebook)	**Description**
30	Roving plant

Area number and description:	3 – Kuipersbult Quarry – surface production
Homogeneous exposure group number	**Description**
1	Crushing and screening plant/s
2	Management, administration, and support services (roving surface)
Activity code (as per DME codebook)	**Description**
20	Crushing
31	Roving surface

Area number and description:	4 – Kuipersbult Quarry – ready-mix plant
Homogeneous exposure group number	**Description**
1	Batching / ready-mix plant
Activity code (as per DME codebook)	**Description**
29	Final product

Area number and description:	5 – Kuipersbult Quarry – engineering maintenance
Homogeneous exposure group number	**Description**
1	Other support services (engineering maintenance / surface workshop/s / roving surface)
Activity code (as per DME codebook)	**Description**
33	Surface workshops

6b Statistical data analysis results for the previous year

Table 1 – airborne pollutants

Area / HEG number	Activity code	Activity description	Pollutant conc. (mg/m³)	Pollutant code	Pollutant conc. %	Pollutant OEL (mg/m³)	Classification band per pollutant
1.1	30	Roving plant		522		0.05	
				459		3.00	
1.2	07	Opencast		522		0.05	
				459		3.00	
2.1	30	Roving plant		522		0.05	
				459		3.00	
3.1	20	Crushing		522		0.05	
				459		3.00	
3.2	31	Roving surface		522		0.05	
				459		3.00	
4.1	29	Final product		522		0.05	
				492		2.00	
5.1	33	Surface workshops		522		0.05	
				459		3.00	

Area / HEG number	Activity code	Activity description	Pollutant conc. (mg/m³)	Pollutant code	Pollutant conc. %	Pollutant OEL (mg/m³)	Classification band per pollutant
				593		0.05	
				459		3.00	
				598		5.00	
				458		10.00	

Table 2 – personal noise

Area / HEG number	**Activity code**	**Activity description**	**Log average personal noise exposure** (dB(A))	**Classification band**
1.1	30	Roving plant		
1.2	07	Opencast		
2.1	30	Roving plant		
3.1	20	Crushing		
3.2	31	Roving surface		
4.1	29	Final product		
5.1	33	Surface workshops		

7 Identified workroom stressors per homogeneous exposure group

7.1	**Area number and description** (as per 6 above):	1 – Kuipersbult Quarry - Opencast
	Activity code and description (as per 6 above):	30 – Roving plant
	Homogeneous exposure group number and description (as per 6 above):	1 - Drilling

Hazardous chemical substances						
Airborne particulate (dust / fume)						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Silica, crystalline (respirable particulate) Quartz (SiO_2) *2 R (**522)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Watering of haul roads (opencast area); and ↘ Drill rig extraction system ↘ Only wet drilling conducted	**Shift:** 07:00-18:00 (11 hrs)	NIOSH 7602	
Route of entry:	Inhalation		**Target organ/s:**	Lungs		
Heath effects:	**Silicosis (inflammation and scaring)** Silica dust particles become trapped in lung tissue causing inflammation and scarring. The particles also reduce the lungs' ability to take in oxygen. This condition is called silicosis. Silicosis results in permanent lung damage and is a progressive, debilitating, and sometimes fatal disease.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Substance	Sources	PPE in use	Current control measures in use	Exposure time in hours	NIOSH method	Category band
Particles not otherwise classified (PNOC) Respirable particulate *2 R (**459)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Watering of haul roads (opencast area); and ↘ Drill rig extraction system ↘ Only wet drilling conducted	**Shift:** 07:00-18:00 (11 hrs)	NIOSH 0600	
Route of entry:	Inhalation		**Target organ/s:**	Lungs		
Heath effects:	**Chronic obstructive pulmonary disease (COPD)** In people with COPD, the lungs can get damaged or clogged with phlegm. Symptoms include cough, sometimes with phlegm, difficulty breathing, wheezing and tiredness.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Gases						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Carbon dioxide (CO_2) *3 Other (**111)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Planned service / maintenance on compressor truck / compressor	**Shift:** 07:00-18:00 (11 hrs)	Discolouration tubes	
Route of entry:	Inhalation		**Target organ/s:**	Respiratory system / Cardiovascular system		
Heath effects:	**Simple Asphyxiant** Exposure to CO_2 can produce a variety of health effects. These may include headaches, dizziness, restlessness, a tingling or needles feeling, difficulty breathing, sweating, tiredness, increased heart rate, elevated blood pressure, coma, asphyxia, and convulsions.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Substance	Sources	PPE in use (Obtain spec sheets)	Current control measures in use	Exposure time in hours	NIOSH method	Category band
Nitric oxide (NO) *3 Other (**437)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Planned service / maintenance on compressor truck / compressor	**Shift:** 07:00-18:00 (11 hrs)	Discolouration tubes	
Route of entry:	Inhalation		**Target organ/s:**	Lungs		
Heath effects:	**Chemical Asphyxiant** Elevated levels of nitrogen dioxide can cause damage to the human respiratory tract and increase a person's vulnerability to, and the severity of, respiratory infections and asthma.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Substance	Sources	PPE in use	Current control measures in use	Exposure time in hours	NIOSH method	Category band
Carbon monoxide (CO) *3 Other (**113)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Planned service / maintenance on compressor truck / compressor	**Shift:** 07:00-18:00 (11 hrs)	Discolouration tubes	
Route of entry:	Inhalation		**Target organ/s:**	Cardiovascular system / Central Nervous System		
Heath effects:	**Carboxyhemoglobin** Breathing air with a high concentration of CO reduces the amount of oxygen that can be transported in the blood stream to critical organs like the heart and brain. At very high levels, which are possible indoors or in other enclosed environments, CO can cause dizziness, confusion, unconsciousness, and death.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Noise				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Refer to Appendix A3 below	Refer to Appendix A3 below	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Noise induced hearing loss.	**Target organ/s:**	Ear (inner ear)	

Thermal: Heat (ambient)				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Natural environmental conditions	Hard hat with brimmed sun protector	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Rapid rises in heat gain due to exposure to hotter than average conditions compromise the body's ability to regulate temperature and can result in a cascade of illnesses, including heat cramps, heat exhaustion, heatstroke, and hyperthermia.			

Thermal: Cold (ambient)				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Natural environmental conditions	Special warm reflector jacket for winter times	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Prolonged exposure to cold will eventually use up your body's stored energy. The result is hypothermia, or abnormally low body temperature. A body temperature that is too low affects the brain, making the victim unable to think clearly or move well. This makes hypothermia particularly dangerous because a person may not know it is happening and will not be able to do anything about it.			

Whole body vibration				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Monitoring frequency**
Refer to Appendix A6 below	Refer to Appendix A6 below	↘ None	**Shift:** 07:00-18:00 (11 hrs)	N/A
Heath effects:	Whole-body vibration can cause fatigue, stomach problems, headache, loss of balance and "shakiness" shortly after or during exposure. The symptoms are like those that many people experience after a long car or boat trip. After daily exposure over several years, whole-body vibration can affect the entire body and result in several health disorders. Studies of bus and truck drivers found that occupational exposure to whole-body vibration could have contributed to several circulatory, bowel, respiratory, muscular, and back disorders. The combined effects of body posture, postural fatigue, dietary habits, and whole-body vibration are the possible causes for these disorders.			

7.2	**Area number and description** (as per 6 above):	1 – Kuipersbult Quarry - Opencast
	Activity code and description (as per 6 above):	07 – Opencast
	Homogeneous exposure group number and description (as per 6 above):	2 - Blasting

Hazardous chemical substances						
Airborne particulate (dust / fume)						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Silica, crystalline (respirable particulate) Quartz (SiO_2) *2 R (**522)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Watering of haul roads (opencast area); and ↘ Drill rig extraction system ↘ Only wet drilling conducted	**Shift:** 07:00-18:00 (11 hrs)	NIOSH 7602	
Route of entry:	Inhalation		**Target organ/s:**	Lungs		
Heath effects:	**Silicosis (inflammation and scaring)** Silica dust particles become trapped in lung tissue causing inflammation and scarring. The particles also reduce the lungs' ability to take in oxygen. This condition is called silicosis. Silicosis results in permanent lung damage and is a progressive, debilitating, and sometimes fatal disease.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Substance	Sources	PPE in use	Current control measures in use	Exposure time in hours	NIOSH method	Category band
Particles not otherwise classified (PNOC) Respirable particulate *2 R (**459)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Watering of haul roads (opencast area); and ↘ Drill rig extraction system ↘ Only wet drilling conducted	**Shift:** 07:00-18:00 (11 hrs)	NIOSH 0600	
Route of entry:	Inhalation		**Target organ/s:**	Lungs		
Heath effects:	**Chronic obstructive pulmonary disease (COPD)** In people with COPD, the lungs can get damaged or clogged with phlegm. Symptoms include cough, sometimes with phlegm, difficulty breathing, wheezing and tiredness.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Noise				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Refer to Appendix A3 below	Refer to Appendix A3 below	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Noise induced hearing loss.	**Target organ/s:**	Ear (inner ear)	

Thermal: Heat (ambient)				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Natural environmental conditions	Hard hat with brimmed sun protector	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Rapid rises in heat gain due to exposure to hotter than average conditions compromise the body's ability to regulate temperature and can result in a cascade of illnesses, including heat cramps, heat exhaustion, heatstroke, and hyperthermia.			

Thermal: Cold (ambient)				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Natural environmental conditions	Special warm reflector jacket for winter times	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Prolonged exposure to cold will eventually use up your body's stored energy. The result is hypothermia, or abnormally low body temperature. A body temperature that is too low affects the brain, making the victim unable to think clearly or move well. This makes hypothermia particularly dangerous because a person may not know it is happening and will not be able to do anything about it.			

7.3	**Area number and description** (as per 6 above):	2 – Kuipersbult Quarry – self-propelled mobile machines
	Activity code and description (as per 6 above):	30 – Roving plant
	Homogeneous exposure group number and description (as per 6 above):	1 – Self-propelled mobile machines

Hazardous chemical substances						
Airborne particulate (dust / fume)						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Silica, crystalline (respirable particulate) Quartz (SiO_2) *2 R (**522)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Watering of haul roads (opencast area); and ↘ Drill rig extraction system ↘ Only wet drilling conducted	**Shift:** 07:00-18:00 (11 hrs)	NIOSH 7602	
Route of entry:	Inhalation		**Target organ/s:**	Lungs		
Heath effects:	**Silicosis (inflammation and scaring)** Silica dust particles become trapped in lung tissue causing inflammation and scarring. The particles also reduce the lungs' ability to take in oxygen. This condition is called silicosis. Silicosis results in permanent lung damage and is a progressive, debilitating, and sometimes fatal disease.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Substance	Sources	PPE in use	Current control measures in use	Exposure time in hours	NIOSH method	Category band
Particles not otherwise classified (PNOC) Respirable particulate *2 R (**459)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Watering of haul roads (opencast area); and ↘ Drill rig extraction system ↘ Only wet drilling conducted	**Shift:** 07:00-18:00 (11 hrs)	NIOSH 0600	
Route of entry:	Inhalation		**Target organ/s:**	Lungs		
Heath effects:	**Chronic obstructive pulmonary disease (COPD)** In people with COPD, the lungs can get damaged or clogged with phlegm. Symptoms include cough, sometimes with phlegm, difficulty breathing, wheezing and tiredness.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Gases						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Carbon dioxide (CO_2) *3 Other (**111)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Planned service / maintenance on compressor truck / compressor	**Shift:** 07:00-18:00 (11 hrs)	Discolouration tubes	
Route of entry:	Inhalation		**Target organ/s:**	Respiratory system / Cardiovascular system		
Heath effects:	**Simple Asphyxiant** Exposure to CO_2 can produce a variety of health effects. These may include headaches, dizziness, restlessness, a tingling or needles feeling, difficulty breathing, sweating, tiredness, increased heart rate, elevated blood pressure, coma, asphyxia, and convulsions.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Substance	Sources	PPE in use (Obtain spec sheets)	Current control measures in use	Exposure time in hours	NIOSH method	Category band
Nitric oxide (NO) *3 Other (**437)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Planned service / maintenance on compressor truck / compressor	**Shift:** 07:00-18:00 (11 hrs)	Discolouration tubes	
Route of entry:	Inhalation		**Target organ/s:**	Lungs		
Heath effects:	**Chemical Asphyxiant** Elevated levels of nitrogen dioxide can cause damage to the human respiratory tract and increase a person's vulnerability to, and the severity of, respiratory infections and asthma.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Substance	Sources	PPE in use	Current control measures in use	Exposure time in hours	NIOSH method	Category band
Carbon monoxide (CO) *3 Other (**113)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Planned service / maintenance on compressor truck / compressor	**Shift:** 07:00-18:00 (11 hrs)	Discolouration tubes	
Route of entry:	Inhalation		**Target organ/s:**	Cardiovascular system / Central Nervous System		
Heath effects:	**Carboxyhemoglobin** Breathing air with a high concentration of CO reduces the amount of oxygen that can be transported in the blood stream to critical organs like the heart and brain. At very high levels, which are possible indoors or in other enclosed environments, CO can cause dizziness, confusion, unconsciousness, and death.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Noise				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Refer to Appendix A3 below	Refer to Appendix A3 below	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Noise induced hearing loss.	**Target organ/s:**	Ear (inner ear)	

Thermal: Heat (ambient)				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Natural environmental conditions	Hard hat with brimmed sun protector	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Rapid rises in heat gain due to exposure to hotter than average conditions compromise the body's ability to regulate temperature and can result in a cascade of illnesses, including heat cramps, heat exhaustion, heatstroke, and hyperthermia.			

Thermal: Cold (ambient)				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Natural environmental conditions	Special warm reflector jacket for winter times	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Prolonged exposure to cold will eventually use up your body's stored energy. The result is hypothermia, or abnormally low body temperature. A body temperature that is too low affects the brain, making the victim unable to think clearly or move well. This makes hypothermia particularly dangerous because a person may not know it is happening and will not be able to do anything about it.			

Whole body vibration				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Monitoring frequency**
Refer to Appendix A6 below	Refer to Appendix A6 below	↘ None	**Shift:** 07:00-18:00 (11 hrs)	N/A
Heath effects:	Whole-body vibration can cause fatigue, stomach problems, headache, loss of balance and "shakiness" shortly after or during exposure. The symptoms are like those that many people experience after a long car or boat trip. After daily exposure over several years, whole-body vibration can affect the entire body and result in several health disorders. Studies of bus and truck drivers found that occupational exposure to whole-body vibration could have contributed to several circulatory, bowel, respiratory, muscular, and back disorders. The combined effects of body posture, postural fatigue, dietary habits, and whole-body vibration are the possible causes for these disorders.			

7.4	**Area number and description** (as per 6 above):	3 – Kuipersbult Quarry – surface production
	Activity code and description (as per 6 above):	30 – Crushing
	Homogeneous exposure group number and description (as per 6 above):	1 – Crushin and screening plant

Hazardous chemical substances						
Airborne particulate (dust / fume)						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Silica, crystalline (respirable particulate) Quartz (SiO_2) *2 R (**522)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Watering of haul roads (opencast area); and ↘ Drill rig extraction system ↘ Only wet drilling conducted	**Shift:** 07:00-18:00 (11 hrs)	NIOSH 7602	
Route of entry:	Inhalation		**Target organ/s:**	Lungs		
Heath effects:	**Silicosis (inflammation and scaring)** Silica dust particles become trapped in lung tissue causing inflammation and scarring. The particles also reduce the lungs' ability to take in oxygen. This condition is called silicosis. Silicosis results in permanent lung damage and is a progressive, debilitating, and sometimes fatal disease.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Substance	Sources	PPE in use	Current control measures in use	Exposure time in hours	NIOSH method	Category band
Particles not otherwise classified (PNOC) Respirable particulate *2 R (**459)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Watering of haul roads (opencast area); and ↘ Drill rig extraction system ↘ Only wet drilling conducted	**Shift:** 07:00-18:00 (11 hrs)	NIOSH 0600	
Route of entry:	Inhalation		**Target organ/s:**	Lungs		
Heath effects:	**Chronic obstructive pulmonary disease (COPD)** In people with COPD, the lungs can get damaged or clogged with phlegm. Symptoms include cough, sometimes with phlegm, difficulty breathing, wheezing and tiredness.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Noise				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Refer to Appendix A3 below	Refer to Appendix A3 below	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Noise induced hearing loss.	**Target organ/s:**	Ear (inner ear)	

Thermal: Heat (ambient)				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Natural environmental conditions	Hard hat with brimmed sun protector	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Rapid rises in heat gain due to exposure to hotter than average conditions compromise the body's ability to regulate temperature and can result in a cascade of illnesses, including heat cramps, heat exhaustion, heatstroke, and hyperthermia.			

Thermal: Cold (ambient)				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Natural environmental conditions	Special warm reflector jacket for winter times	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Prolonged exposure to cold will eventually use up your body's stored energy. The result is hypothermia, or abnormally low body temperature. A body temperature that is too low affects the brain, making the victim unable to think clearly or move well. This makes hypothermia particularly dangerous because a person may not know it is happening and will not be able to do anything about it.			

Whole body vibration				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Monitoring frequency**
Refer to Appendix A6 below	Refer to Appendix A6 below	↘ None	**Shift:** 07:00-18:00 (11 hrs)	N/A
Heath effects:	Whole-body vibration can cause fatigue, stomach problems, headache, loss of balance and "shakiness" shortly after or during exposure. The symptoms are like those that many people experience after a long car or boat trip. After daily exposure over several years, whole-body vibration can affect the entire body and result in several health disorders. Studies of bus and truck drivers found that occupational exposure to whole-body vibration could have contributed to several circulatory, bowel, respiratory, muscular, and back disorders. The combined effects of body posture, postural fatigue, dietary habits, and whole-body vibration are the possible causes for these disorders.			

7.5	**Area number and description** (as per 6 above):	3 – Kuipersbult Quarry – surface production
	Activity code and description (as per 6 above):	30 – Roving plant
	Homogeneous exposure group number and description (as per 6 above):	2 – Management, administration, and supoort services

Hazardous chemical substances						
Airborne particulate (dust / fume)						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Silica, crystalline (respirable particulate) Quartz (SiO_2) *2 R (**522)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Watering of haul roads (opencast area); and ↘ Drill rig extraction system ↘ Only wet drilling conducted	**Shift:** 07:00-18:00 (11 hrs)	NIOSH 7602	
Route of entry:	Inhalation		**Target organ/s:**	Lungs		
Heath effects:	**Silicosis (inflammation and scaring)** Silica dust particles become trapped in lung tissue causing inflammation and scarring. The particles also reduce the lungs' ability to take in oxygen. This condition is called silicosis. Silicosis results in permanent lung damage and is a progressive, debilitating, and sometimes fatal disease.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Substance	Sources	PPE in use	Current control measures in use	Exposure time in hours	NIOSH method	Category band
Particles not otherwise classified (PNOC) Respirable particulate *2 R (**459)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Watering of haul roads (opencast area); and ↘ Drill rig extraction system ↘ Only wet drilling conducted	**Shift:** 07:00-18:00 (11 hrs)	NIOSH 0600	
Route of entry:	Inhalation		**Target organ/s:**	Lungs		
Heath effects:	**Chronic obstructive pulmonary disease (COPD)** In people with COPD, the lungs can get damaged or clogged with phlegm. Symptoms include cough, sometimes with phlegm, difficulty breathing, wheezing and tiredness.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Noise				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Refer to Appendix A3 below	Refer to Appendix A3 below	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Noise induced hearing loss.	**Target organ/s:**	Ear (inner ear)	

Thermal: Heat (ambient)				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Natural environmental conditions	Hard hat with brimmed sun protector	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Rapid rises in heat gain due to exposure to hotter than average conditions compromise the body's ability to regulate temperature and can result in a cascade of illnesses, including heat cramps, heat exhaustion, heatstroke, and hyperthermia.			

Thermal: Cold (ambient)				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Natural environmental conditions	Special warm reflector jacket for winter times	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Prolonged exposure to cold will eventually use up your body's stored energy. The result is hypothermia, or abnormally low body temperature. A body temperature that is too low affects the brain, making the victim unable to think clearly or move well. This makes hypothermia particularly dangerous because a person may not know it is happening and will not be able to do anything about it.			

7.6	**Area number and description** (as per 6 above):	4 – Kuipersbult Quarry – ready-mix
	Activity code and description (as per 6 above):	29 – final product
	Homogeneous exposure group number and description (as per 6 above):	1 – Ready-mix / batching plant

Hazardous chemical substances						
Airborne particulate (dust / fume)						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Silica, crystalline (respirable particulate) Quartz (SiO_2) *2 R (**522)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Watering of haul roads (opencast area); and ↘ Drill rig extraction system ↘ Only wet drilling conducted	**Shift:** 07:00-18:00 (11 hrs)	NIOSH 7602	
Route of entry:	Inhalation		**Target organ/s:**	Lungs		
Heath effects:	**Silicosis (inflammation and scaring)** Silica dust particles become trapped in lung tissue causing inflammation and scarring. The particles also reduce the lungs' ability to take in oxygen. This condition is called silicosis. Silicosis results in permanent lung damage and is a progressive, debilitating, and sometimes fatal disease.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Substance	Sources	PPE in use	Current control measures in use	Exposure time in hours	NIOSH method	Category band
Particles not otherwise classified (PNOC) Respirable particulate *2 R (**459)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Watering of haul roads (opencast area); and ↘ Drill rig extraction system ↘ Only wet drilling conducted	**Shift:** 07:00-18:00 (11 hrs)	NIOSH 0600	
Route of entry:	Inhalation		**Target organ/s:**	Lungs		
Heath effects:	**Chronic obstructive pulmonary disease (COPD)** In people with COPD, the lungs can get damaged or clogged with phlegm. Symptoms include cough, sometimes with phlegm, difficulty breathing, wheezing and tiredness.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Noise				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Refer to Appendix A3 below	Refer to Appendix A3 below	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Noise induced hearing loss.	**Target organ/s:**	Ear (inner ear)	

Thermal: Heat (ambient)				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Natural environmental conditions	Hard hat with brimmed sun protector	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Rapid rises in heat gain due to exposure to hotter than average conditions compromise the body's ability to regulate temperature and can result in a cascade of illnesses, including heat cramps, heat exhaustion, heatstroke, and hyperthermia.			

Thermal: Cold (ambient)				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Natural environmental conditions	Special warm reflector jacket for winter times	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Prolonged exposure to cold will eventually use up your body's stored energy. The result is hypothermia, or abnormally low body temperature. A body temperature that is too low affects the brain, making the victim unable to think clearly or move well. This makes hypothermia particularly dangerous because a person may not know it is happening and will not be able to do anything about it.			

7.7	**Area number and description** (as per 6 above):	5 – Kuipersbult Quarry – engineering maintenance
	Activity code and description (as per 6 above):	33 – Surface workshops
	Homogeneous exposure group number and description (as per 6 above):	1 – Engineering maintenance / workshop/s

Hazardous chemical substances						
Airborne particulate (dust / fume)						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Silica, crystalline (respirable particulate) Quartz (SiO_2) *2 R (**522)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Watering of haul roads (opencast area); and ↘ Drill rig extraction system ↘ Only wet drilling conducted	**Shift:** 07:00-18:00 (11 hrs)	NIOSH 7602	
Route of entry:	Inhalation		**Target organ/s:**	Lungs		
Heath effects:	**Silicosis (inflammation and scaring)** Silica dust particles become trapped in lung tissue causing inflammation and scarring. The particles also reduce the lungs' ability to take in oxygen. This condition is called silicosis. Silicosis results in permanent lung damage and is a progressive, debilitating, and sometimes fatal disease.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Substance	Sources	PPE in use	Current control measures in use	Exposure time in hours	NIOSH method	Category band
Particles not otherwise classified (PNOC) Respirable particulate *2 R (**459)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Watering of haul roads (opencast area); and ↘ Drill rig extraction system ↘ Only wet drilling conducted	**Shift:** 07:00-18:00 (11 hrs)	NIOSH 0600	
Route of entry:	Inhalation		**Target organ/s:**	Lungs		
Heath effects:	**Chronic obstructive pulmonary disease (COPD)** In people with COPD, the lungs can get damaged or clogged with phlegm. Symptoms include cough, sometimes with phlegm, difficulty breathing, wheezing and tiredness.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Hazardous chemical substances						
Airborne particulate (dust / fumes)						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Welding fumes						
Iron oxide, dust & fumes (Fe_2O_3) as Fe *1 T (**348)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Welding done in open area and not confined area / workshop/s; ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** vanadium on surface being welded; and ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** paint, grease, oil, etc. on surface being welded.	**Shift:** 07:00-18:00 <30 minutes / 11 hrs shift	NIOSH 7300	
Route of entry:	Inhalation		**Target organ/s:**	Respiratory system		
Heath effects:	**Metal Fume Fever** Exposure to Iron Oxide (Fe_2O_3) fumes can cause metal fume fever. This is a flu-like illness with symptoms of metallic taste, fever and chills, aches, chest tightness and cough. Prolonged or repeated contact can discolor the eyes causing permanent Iron staining.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Hazardous chemical substances						
Airborne particulate (dust / fumes)						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Aluminum welding fumes as Al *1 T (**20A)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Welding done in open area and not confined area / workshop/s; ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** vanadium on surface being welded; and ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** paint, grease, oil, etc. on surface being welded.	**Shift:** 07:00-18:00 <30 minutes / 11 hrs shift	NIOSH 7300	
Route of entry:	Inhalation		**Target organ/s:**	Respiratory system		
Heath effects:	**Metal Fume Fever** Exposure to Iron Oxide (Fe_2O_3) fumes can cause metal fume fever. This is a flu-like illness with symptoms of metallic taste, fever and chills, aches, chest tightness and cough. Prolonged or repeated contact can discolor the eyes causing permanent Iron staining.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Hazardous chemical substances						
Airborne particulate (dust / fumes)						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Manganese fume as Mn *1 T (**379)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Welding done in open area and not confined area / workshop/s; ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** vanadium on surface being welded; and ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** paint, grease, oil, etc. on surface being welded.	**Shift:** 07:00-18:00 <30 minutes / 11 hrs shift	NIOSH 7300	
Route of entry:	Inhalation		**Target organ/s:**	Respiratory system		
Heath effects:	**Metal Fume Fever** Magnesium oxide fumes may cause irritation of the eyes and nose. It may also cause metal fume fever. Symptoms of metal fume fever include chills, fever, headache, tightness of chest, cough, weakness, tiredness, dryness of the nose and mouth, muscular pain, nausea, and vomiting.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Hazardous chemical substances						
Airborne particulate (dust / fumes)						
Substance	**Sources**	**PPE in use** sheets)	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Copper fume as Cu *1 T (**154)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Welding done in open area and not confined area / workshop/s; ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** vanadium on surface being welded; and ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** paint, grease, oil, etc. on surface being welded.	**Shift:** 07:00-18:00 <30 minutes / 11 hrs shift	NIOSH 7300	
Route of entry:	Inhalation		**Target organ/s:**	Respiratory system		
Heath effects:	**Metal Fume Fever** Exposure to copper fumes cause upper respiratory tract irritation, metallic taste, nausea, and metal fume fever.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Hazardous chemical substances						
Airborne particulate (dust / fumes)						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Molybdenum compounds as Mo *1 T (**422/423)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Welding done in open area and not confined area / workshop/s; ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** vanadium on surface being welded; and ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** paint, grease, oil, etc. on surface being welded.	**Shift:** 07:00-18:00 <30 minutes / 11 hrs shift	NIOSH 7300	
Route of entry:	Inhalation		**Target organ/s:**	Eyes, respiratory system, liver & kidneys		
Heath effects:	**Acute effects are eye, nose and throat irritation, and shortness of breath** Exposure to Molybdenum can cause headache, fatigue, loss of appetite, muscle and joint pain. Repeated exposure may raise the Uric Acid level in the body, which can lead to gout. Molybdenum may damage the liver and kidneys.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Hazardous chemical substances						
Airborne particulate (dust / fumes)						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Vanadium fume and respirable particulate (V_2O_5) as V *2 R (**593)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Welding done in open area and not confined area / workshop/s; ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** vanadium on surface being welded; and ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** paint, grease, oil, etc. on surface being welded.	**Shift:** 07:00-18:00 <30 minutes / 11 hrs shift	NIOSH 7504	
Route of entry:	Inhalation		**Target organ/s:**	Respiratory system		
Heath effects:	**Potential Carcinogen** Exposure to vanadium dust can result in respiratory tract irritation. Bronchospasms and bronchitis, accompanied by wheezing and dyspnea may occur, and bronchopneumonia may result from very high and prolonged exposures. The mechanism of vanadium's effect on the respiratory system is like that of other metals. Vanadium damages alveolar macrophages and affects lung clearance rates.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Hazardous chemical substances						
Airborne particulate (dust / fumes)						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Zinc oxide, fume (Zn O) as Zn *1 T (**606)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Welding done in open area and not confined area / workshop/s; ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** vanadium on surface being welded; and ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** paint, grease, oil, etc. on surface being welded.	**Shift:** 07:00-18:00 <30 minutes / 11 hrs shift	NIOSH 7300	
Route of entry:	Inhalation		**Target organ/s:**	Respiratory system		
Heath effects:	**Metal Fume Fever** Exposure to Zinc oxide can cause "metal fume fever." This is a flu-like illness with symptoms of metallic taste in the mouth, headache, fever and chills, aches, chest tightness and cough. The symptoms may be delayed for several hours after exposure and usually last for a day or two.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Hazardous chemical substances						
Airborne particulate (dust / fumes)						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Nickel, inorganic compounds as Ni *1 T (**432/433)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Welding done in open area and not confined area / workshop/s; ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** vanadium on surface being welded; and ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** paint, grease, oil, etc. on surface being welded.	**Shift:** 07:00-18:00 <30 minutes / 11 hrs shift	NIOSH 7300	
Route of entry:	Inhalation		**Target organ/s:**	Nervous system		
Heath effects:	**Neurotoxin** Contact dermatitis; headaches; gastrointestinal manifestations; respiratory manifestations; lung fibrosis; cardiovascular diseases; lung cancer; nasal cancer; epigenetic effects.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Hazardous chemical substances						
Airborne particulate (dust / fumes)						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Chromium (II) compounds as Cr *1 T (**147)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Welding done in open area and not confined area / workshop/s; ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** vanadium on surface being welded; and ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** paint, grease, oil, etc. on surface being welded.	**Shift:** 07:00-18:00 <30 minutes / 11 hrs shift	NIOSH 7300	
Route of entry:	Inhalation		**Target organ/s:**	Respiratory system		
Heath effects:	**Carcinogen** Human occupational experience clearly indicates that, when inhaled, chromium compounds are respiratory tract irritants, resulting in airway irritation, airway obstruction, lungs, nasal, or sinus cancer.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Hazardous chemical substances						
Airborne particulate (dust / fumes)						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Cobalt and cobalt compounds as Co *1 T (**153)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Welding done in open area and not confined area / workshop/s; ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** vanadium on surface being welded; and ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** paint, grease, oil, etc. on surface being welded.	**Shift:** 07:00-18:00 <30 minutes / 11 hrs shift	NIOSH 7300	
Route of entry:	Inhalation		**Target organ/s:**	Respiratory system		
Heath effects:	**Potential / Possible Carcinogen** Cobalt itself may cause allergic dermatitis, rhinitis, and asthma. Specific IgE against a complex of cobalt with albumin can sometimes be shown and a bronchial provocation test with a cobalt salt may be positive. Inhalation of cobalt containing dust has also led to pathologic reactions in the lung parenchyma. The lesions, called 'hard metal disease', have ranged from intense alveolitis resembling desquamative or giant-cell interstitial pneumonitis to end-stage pulmonary fibrosis.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Hazardous chemical substances						
Airborne particulate (dust / fumes)						
Substance	**Sources**	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**NIOSH method**	**Category band**
Titanium dioxide (TiO_2) as Ti *1 T (**569)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Welding done in open area and not confined area / workshop/s; ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** vanadium on surface being welded; and ↘ A special instruction and procedure for removing (cleaning) all surface/s to be welded to ensure **NO** paint, grease, oil, etc. on surface being welded.	**Shift:** 07:00-18:00 <30 minutes / 11 hrs shift	NIOSH 7300	
Route of entry:	Inhalation		**Target organ/s:**	Respiratory system		
Heath effects:	**Potential / Possible Carcinogen** Note that titanium dioxide has been classified by the American Conference of Governmental Industrial Hygienists (ACGIH) as A3 – Confirmed animal carcinogen, and by the International Agency for Research on Cancer (IARC) as 2B - Possibly carcinogenic to humans.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Substance	Sources	PPE in use	Current control measures in use	Exposure time in hours	NIOSH method	Category band
Ozone (O_2) *3 Other (**454)	Refer to Appendix A2.1 below	Refer to Appendix A2.1 below	↘ Planned service / maintenance on compressor truck / compressor	**Shift:** 07:00-18:00 (11 hrs)	Discolouration tubes	
Route of entry:	Inhalation		**Target organ/s:**	Cardiovascular system / Central Nervous System		
Heath effects:	**Carboxyhemoglobin** Breathing air with a high concentration of CO reduces the amount of oxygen that can be transported in the blood stream to critical organs like the heart and brain. At very high levels, which are possible indoors or in other enclosed environments, CO can cause dizziness, confusion, unconsciousness, and death.					

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Noise				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Refer to Appendix A3 below	Refer to Appendix A3 below	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Noise induced hearing loss.	**Target organ/s:**	Ear (inner ear)	

Thermal: Heat (artificial)				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Refer to Appendix A4 below	Refer to Appendix A4 below	↘ N/A	N/A	
Heath effects:	Rapid rises in heat gain due to exposure to hotter than average conditions compromise the body's ability to regulate temperature and can result in a cascade of illnesses, including heat cramps, heat exhaustion, heatstroke, and hyperthermia.			

Thermal: Cold (artificial)				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Refer to Appendix A5 below	Refer to Appendix A5 below	↘ N/A	N/A	
Heath effects:	Prolonged exposure to cold will eventually use up your body's stored energy. The result is hypothermia, or abnormally low body temperature. A body temperature that is too low affects the brain, making the victim unable to think clearly or move well. This makes hypothermia particularly dangerous because a person may not know it is happening and will not be able to do anything about it.			

Thermal: Heat (ambient)				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Natural environmental conditions	Hard hat with brimmed sun protector	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Rapid rises in heat gain due to exposure to hotter than average conditions compromise the body's ability to regulate temperature and can result in a cascade of illnesses, including heat cramps, heat exhaustion, heatstroke, and hyperthermia.			

Thermal: Cold (ambient)				
Sources	**PPE in use**	**Current control measures in use**	**Exposure time in hours**	**Category band**
Natural environmental conditions	Special warm reflector jacket for winter times	↘ None	**Shift:** 07:00-18:00 (11 hrs)	
Heath effects:	Prolonged exposure to cold will eventually use up your body's stored energy. The result is hypothermia, or abnormally low body temperature. A body temperature that is too low affects the brain, making the victim unable to think clearly or move well. This makes hypothermia particularly dangerous because a person may not know it is happening and will not be able to do anything about it.			

8 Mechanisation

The operation is ±80% mechanised with approximately 20% manual support.

- Opencast – drilling and blasting: -
  - Drill rig/s.

- Self-propelled mobile machines (roving plant): -
  - Excavators;
  - Articulated dump trucks;
  - Front end loaders;
  - Bulldozer;
  - Several light delivery vehicles; and
  - Water bowser / trailer.

- Crushing and screening plant (crushing): -
  - Primary crusher plant with several conveyors;
  - Secondary crusher plant with several conveyors; and
  - Screens with several conveyors.

- Management, administration, and support services (roving surface):
  - Management tasks and support services tasks.

- Engineering maintenance (surface workshop/s):
  - Generator/s;
  - Compressor/s;
  - Welders;
  - Electrical power;
  - Cutting touch;
  - Grinders; and
  - Contracted engineering tasks with relevant equipment.

Refer to asset register for quantities and full equipment details.

9 General ventilation arrangements

This is a surface operation only and except for general surface ventilation arrangements, such as in buildings, no other ventilation arrangements are applicable.

10 All statutory mandatory codes of practices

- Mandatory code of practice for the safe use of conveyor belt installations for the transportation of minerals, material, or personnel;
- Mandatory code of practice for an occupational health programme on personal exposure to airborne pollutants;
- Mandatory code of practice for an occupational health programme for personal exposure to noise;
- Mandatory code of practice for an occupational health programme for thermal stress;
- Mandatory code of practice for the prevention, mitigation, and management of covid – 19 outbreak;
- Mandatory code of practice for minimum standards of fitness;
- Mandatory code of practice for the management of medical incapacity due to ill-health and injury;
- Mandatory code of practice for risk-based emergency medical care on a mine;
- Mandatory code of practice for the provision for PPE for women in mines;
- Mandatory code of practice for the prevention of fires at mines;
- Mandatory code of practice for emergency preparedness and response;
- Mandatory code of practice for trackless mobile machines;
- Mandatory code of practice for risk-based fatigue management at mines;
- Mandatory code of practice for management of working in confined spaces in mines;
- Mandatory code of practice for a quality assurance programme for a system of occupational hygiene and ventilation engineering measurements
- Mandatory code of practice for the right to refuse dangerous work and leave a dangerous working place.

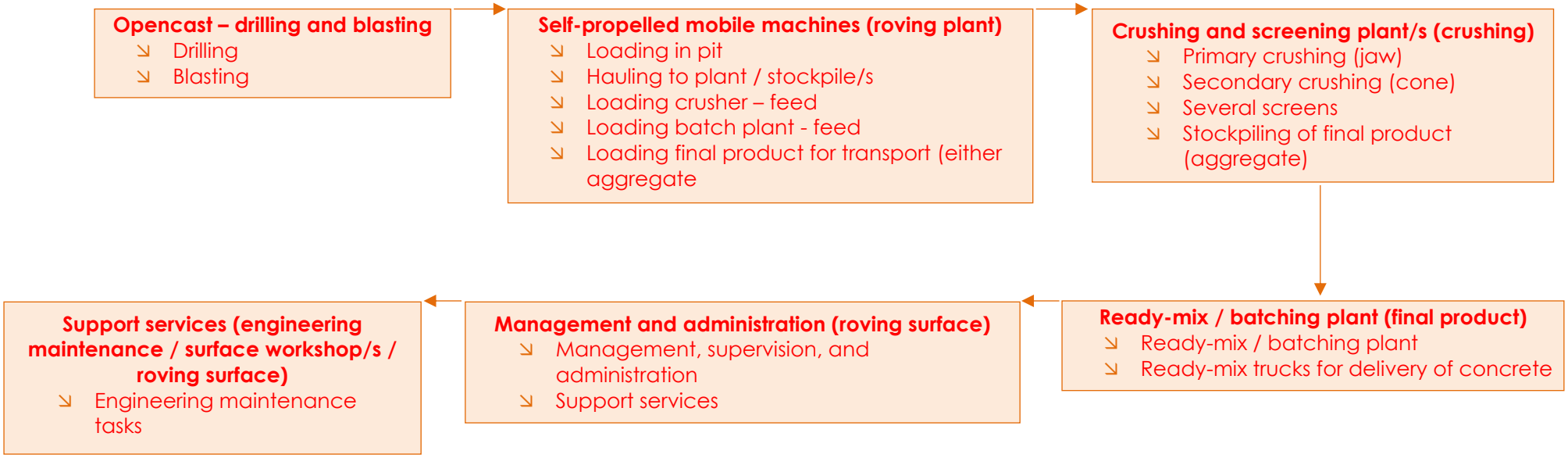
All COPs are managed in an integrated management system.

11 Unique features on the mine

No industry specific unique features.

Appendix A1
Organisation structure / organogram

Appendix A2
Process flow diagram



Appendix A2.1
List of all hazardous chemical substance sources per area and homogeneous exposure group and required PPE

Area and homogeneous exposure group number		**Homogeneous exposure group description**	
1.5		Engineering maintenance / surface workshop/s (roving surface)	
System number	**Substance (NB! Same as in table *8.1.1 - 8.1.2 …* above)**	**Source / process description**	**Exposed occupations (NB As per client description) and required PPE**
1	Silica, crystalline (respirable particulate) Quartz (SiO_2) *2 R (**522)	**Opencast** ↘ Drill rig - drilling.	↘ Refer to the latest labour complement.
2	Particles not otherwise classified (PNOC) Respirable particulate *2 R (**459)		
3	Iron oxide, dust & fumes (Fe_2O_3) as Fe *1 T (**348)	**Arc welding** ↘ Generator/s; and ↘ Welder/s.	
4	Aluminum welding fumes as Al *1 T (**20A)		
5	Manganese fume as Mn *1 T (**379)		
6	Copper fume as Cu *1 T (**154)		

Area and homogeneous exposure group number		**Homogeneous exposure group description**	
1.5		Engineering maintenance / surface workshop/s (roving surface)	
System number	**Substance (NB! Same as in table *8.1.1 - 8.1.2 …* above)**	**Source / process description**	**Exposed occupations (NB As per client description) and required PPE**
7	Molybdenum compounds as Mo *1 T (**422/423)		
8	Vanadium fume and respirable particulate (V_2O_5) as V *1 T (**593)		
9	Zinc oxide, fume (Zn O) as Zn *1 T (**606)		
10	Nickel, inorganic compounds as Ni *1 T (**432/433)		
11	Chromium (II) compounds as Cr *1 T (**147)		
12	Cobalt and cobalt compounds as Co *2 R (**153)		
13	Titanium dioxide (Ti O_2) as Ti		

Area and homogeneous exposure group number		**Homogeneous exposure group description**	
1.5		Engineering maintenance / surface workshop/s (roving surface)	
System number	**Substance (NB! Same as in table *8.1.1 - 8.1.2 …* above)**	**Source / process description**	**Exposed occupations (NB As per client description) and required PPE**
	*1 T (**569)		
14	Carbon dioxide (CO_2) *3 Other (**111)		
15	Nitric oxide (NO) *3 Other (**437)		
16	Carbon monoxide (CO) *3 Other (**113)		
17	Ozone (O_3) *3 Other (**454)		
18	Particles not otherwise classified (PNOC) Inhalable particulate *2 T (**485)		

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Refer to latest electronic copy of hazardous chemical substance register for required PPE!

Appendix A2.2
List of all respirator zones per area and homogeneous exposure group

Area and homogeneous exposure group number		**Homogeneous exposure group description**	
1.5		Engineering maintenance / surface workshop/s (roving surface)	
System number	**Substance (NB! Same as in table *8.1.1 - 8.1.2 …* above)**	**Respirator zone description**	**Exposed occupations (NB As per client description) and required PPE**
1	Silica, crystalline (respirable particulate) Quartz (SiO_2) *2 R (**522)	↘ Various location where engineering maintenance tasks are performed.	↘ Refer to the latest electronic copy of the labour complement.
2	Particles not otherwise classified (PNOC) Respirable particulate *2 R (**459)		
3	Iron oxide, dust & fumes ($Fe_2 O_3$) as Fe *1 T (**348)		
4	Aluminum welding fumes as Al *1 T (**20A)		
5	Manganese fume as Mn *1 T (**379)		
6	Copper fume as Cu *1 T (**154)		

Area and homogeneous exposure group number		**Homogeneous exposure group description**	
1.5		Engineering maintenance / surface workshop/s (roving surface)	
System number	**Substance (NB! Same as in table *8.1.1 - 8.1.2 …* above)**	**Respirator zone description**	**Exposed occupations (NB As per client description) and required PPE**
7	Molybdenum compounds as Mo *1 T (**422/423)		
8	Vanadium fume and respirable particulate (V_2O_5) as V *1 T (**593)		
9	Zinc oxide, fume (Zn O) as Zn *1 T (**606)		
10	Nickel, inorganic compounds as Ni *1 T (**432/433)		
11	Chromium (II) compounds as Cr *1 T (**147)		
12	Cobalt and cobalt compounds as Co *1 T (**153)		
13	Titanium dioxide (Ti O_2) as Ti		

Area and homogeneous exposure group number		**Homogeneous exposure group description**	
1.5		Engineering maintenance / surface workshop/s (roving surface)	
System number	**Substance (NB! Same as in table *8.1.1 - 8.1.2 …* above)**	**Respirator zone description**	**Exposed occupations (NB As per client description) and required PPE**
	*1 T (**569)		
	Particles not otherwise classified (PNOC) Respirable particulate *1 T (**485)		

*1 = Total sample *2 = Respirable sample *3 = Other ** = DMR pollutant code

Appendix A3
List of all noise sources (Table 1), zones (Table 2) with plans, per area and homogeneous exposure group and required PPE

Table 1:

Area and homogeneous exposure group number	**Homogeneous exposure group description**	
1.1	Opencast – drilling and blasting (contract drilling & blasting – *from time to time, not full time*)	
System number	**Noise source description**	**Exposed occupations (NB As per client description) and PPE**
1	↘ Drill rig.	↘ Drill operator / supervisor (ear plugs); and ↘ Drill operator assistant (ear plugs).
2	↘ Drill rig compressor truck.	↘ Drill operator / supervisor (ear plugs); and ↘ Drill operator assistant (ear plugs).
3	↘ Drill rig compressor.	↘ Drill operator / supervisor (ear plugs); and ↘ Drill operator assistant (ear plugs).

Refer to latest electronic copy of the noise source register for individual equipment details!

Table 2:

Area and homogeneous exposure group number	**Homogeneous exposure group description**	
1.1	Opencast – drilling and blasting (contract drilling & blasting – *from time to time, not full time*)	
System number	**Noise zone description**	**Exposed occupations (NB As per client description)**
1	↘ Drill rig.	↘ Drill operator / supervisor; and ↘ Drill operator assistant.
2	↘ Drill rig compressor truck.	↘ Drill operator / supervisor; and ↘ Drill operator assistant.
3	↘ Drill rig compressor.	↘ Drill operator / supervisor; and ↘ Drill operator assistant.

Refer to latest electronic copy of the noise source register for noise zone plans!

Appendix A4
List of all artificial heat sources per area and homogeneous exposure group and required PPE

Area and homogeneous exposure group number	**Homogeneous exposure group description**	
1.5	Engineering maintenance / surface workshop/s (roving surface)	
System number	**Artificial heat sources**	**Exposed occupations (NB As per client description) and PPE**
1	↘ Welding and flame cutting.	↘ Artisan / Assistant (welding helmet, gloves and apron)

Refer to latest electronic copy of the artificial thermal source register for full details and required PPE!

Appendix A5
List of all artificial cold sources per area and homogeneous exposure group and required PPE

Area and homogeneous exposure group number	**Homogeneous exposure group description**	
1.1	Opencast – drilling and blasting (contract drilling & blasting – *from time to time, not full time*)	
System number	**Artificial cold sources**	**Exposed occupations (NB As per client description) and required PPE**
1	↘ None	↘ N/A

Refer to latest electronic copy of the artificial thermal source register for full details and required PPE!

Appendix A6
List of all vibration sources per area and homogeneous exposure group and required PPE

Area and homogeneous exposure group number	Homogeneous exposure group description	
1.2	Self-propelled mobile machines (roving plant)	
System number	**Source and process description**	**Exposed occupations (NB As per client description) and PPE**
1	↘ Excavators.	↘ Operator – Excavators (None)
3	↘ Articulated dump trucks.	↘ Operator – Dumper trucks (None)
5	↘ Front end loaders.	↘ Operator – Front end loader (None)
7	↘ Bulldozer.	↘ Operator – Bulldozer (None)
8	↘ Water truck.	↘ Operator – Water truck (None)

Appendix A7

Total labour compliment (including contractors) per area and homogeneous exposure group with readable copies of their respective identity documents / passports

Refer to latest electronic copy of the labour complement for full details!

Appendix A8.1
Emergency preparedness and response layout plan with all relevant information

Appendix A8.2
Emergency preparedness and response emergency contact numbers

Emergency service description	Contact person if available / applicable	Contact number
		+27 (0)
		+27 (0)
		+27 (0)
		+27 (0)
		+27 (0)
		+27 (0)
		+27 (0)
		+27 (0)

Appendix A8.3
Emergency preparedness and response – map showing nearest route / police station

Lephalale town

Marapong mobile station

Appendix A8.4
Emergency preparedness and response – map showing nearest route / ambulance

Lephale control centre

Marapong private hospital

Mediclinic Lephalale

Ellisras district hospital

Appendix A8.5
Emergency preparedness and response – map showing nearest route / doctor

Dr Seropa surgery Lephalale

Appendix A8.6
Emergency preparedness and response – map showing nearest route / fire brigade / station

Lephalale fire and disater management centre

Appendix A8.7

Emergency preparedness and response – map showing nearest route / hospital

Appendix A8.8
Emergency preparedness and response – traffic management plan details

Appendix A8.9
Emergency preparedness and response – firefighting equipment details

Number	Extinguisher number	Type and size	Location / description
1	VTM01	Dry Chemical Powder/9KG/STD	Admin Office
2	VTM02	Dry Chemical Powder/9KG/STD	Geologist Office
3	VTM03	Dry Chemical Powder/9KG/STD	Water Truck
4	VTM005	Dry Chemical Powder/9KG/STD	Diesel Bowser
5	VTM004	Dry Chemical Powder/250KG/STD	Diesel Bowser (depot)
6	VTM003	Dry Chemical Powder/250KG/STD	Diesel Bowser (depot)
7	VTM014	Dry Chemical Powder/9KG/STD	SANY Excavator MAJ008
8	VTM050	Dry Chemical Powder/9KG/STD	Komatsu HF04
9	VTM016	Dry Chemical Powder/9KG/STD	Kobelco Excavator HF78
10	VTM017	Dry Chemical Powder/9KG/STD	Beu ADT HF67
11	VTM22	Dry Chemical Powder/9KG/STD	Beu ADT HF053
12	VTM007	Dry Chemical Powder/9KG/STD	AMC Tools Container
13	AMC008	Dry Chemical Powder/9KG/STD	AMC Boiler Maker Store - AMC008
14	AMC006	Dry Chemical Powder/9KG/STD	Behind Store Room - AMC006
15	AMC015	Dry Chemical Powder/9KG/STD	AMC Fatigue Room - AMC015
16	VTM014	Dry Chemical Powder/9KG/STD	AMC Gen 002 - AMC005
17	AMC001	Dry Chemical Powder/9KG/STD	AMC Gen Exit - AMC001
18	VTM024	Dry Chemical Powder/9KG/STD	AMC Electrical DB Box - AMC029

Appendix A8.10
Emergency preparedness and response – first aid boxes details

Number	First aid box number	Location / description

Appendix A8.11
Emergency preparedness and response – fire hazards register

Number	Fire hazard description	Fire hazard class	Fuel Source	Ignition source	Fixed / mobile	Location

Annexure A
List of all hazardous chemical substance used in HEG / stored / kept on site

HEG number	Name of hazardous chemical substance (SDS)	Manufacturer	Supplier	Quantities used / month	Quantities stored / month

For most updated details and safety data sheet, refer to electronic version of hazardous chemical substance register.

Annexure B
Inventory of trackless mobile machines on site

Make / Manufacturer	Fleet number	Model (other specs)	Description	Power output (kW)
Komatsu	HF 063	B30D	Dump truck	
Bell	HF 053	B30	Dump truck	
Bell	NTSAD	B45	Dump truck	
Bell	NTSADT	B40	Dump truck	
Hitachi			Front end loader	
Komatsu	AMC 008	WA008	Front end loader	
Hyundai	S 05	220 LC95	Excavator	
Volvo	MPH 002		Excavator	
Dosan	CV 001		Excavator	
Sany			Excavator	
Bell	HF 78	380D	Excavator	
Hino			Water tanker	

Annexure C
Trained and certificated first aiders and designated area of responsibility

WFH number	Initials and surname	Certificate expiry date	Designated area of responsibility	Appointment number

Annexure D

Trained and certificated fire fighters and designated area of responsibility

WFH number	Initials and surname	Certificate expiry date	Designated area of responsibility	Appointment number

Annexure E
Statistical data analysis for the previous year, with graphs (airborne pollutants)

Note
The mine standard for statistical analysis is based on the 95th percentile, but for legal reporting the 90th percentile is used. This is done to ensure that legal reporting is within required limits.

Statitical anaysis is updated on an ongoing basis and this latest anaysis in only available in electronic version.

Annexure F
Statistical data analysis for the previous year, with graphs (noise)

Note

The mine standard for statistical analysis is based on the 95th percentile, but for legal reporting the 90th percentile is used. This is done to ensure that legal reporting is within required limits.

Statitical anaysis is updated on an ongoing basis and this latest anaysis in only available in electronic version.

Attachment A1.1

Certificate issued by the Commissioner of Companies & Intellectual Property Commission on Thursday, October 5, 2023 at 9:48

Companies and Intellectual Property Commission
a member of the dti group

Disclosure Certificate: Companies and Close Corporations

Registration Number: 2002 / 025895 / 07
Enterprise Name: AFRIMAT CONTRACTING INTERNATIONAL

ENTERPRISE INFORMATION

Registration Number	2002 / 025895 / 07
Enterprise Name	AFRIMAT CONTRACTING INTERNATIONAL (PTY) LTD
Registration Date	18/10/2002
Business Start Date	18/10/2002
Enterprise Type	Private Company
Enterprise Status	In Business
Compliance Notice Status	NONE
Financial Year End	February
TAX Number	9150977149

Addresses	POSTAL ADDRESS	ADDRESS OF REGISTERED OFFICE
	P O BOX 505 WORCESTER WESTERN CAPE 6849	3 SENNETT STREET WORCESTER WESTERN CAPE 0850

ACTIVE MEMBERS / DIRECTORS

Surname and First Names	Type	ID Number / Date of Birth	Contrib. (R)	Interest (%)	Appoint. Date	Address
AFRIMAT,	Secretary (Companies and CC's)	1981-01-11	0.00	0.00	02/09/2022	Postal: P O BOX 5278, TYGER VALLEY, 7536 Residential:
TEFFO, ESTER FIKILE	Director	8707140335086	0.00	0.00	01/01/2016	Postal: P O BOX 14014, LYTTELTON, LYTTELTON, GAUTENG, 0140 Residential: 275 WEST STREET, UNIT 60 THE OVAL, CENTURION, GAUTENG, 0157
GILES, DAVIN VINTCENT	Director	6902025200081	0.00	0.00	04/07/2023	Postal: 12 DE HOOP CLOSE, CLARA ANNA FONTEIN, DURBANVILLE, WESTERN CAPE, Residential: 12 DE HOOP CLOSE, CLARA ANNA FONTEIN, DURBANVILLE, WESTERN CAPE,
DE WIT, PIETER GABRIEL STEPHANUS	Director	7307315059081	0.00	0.00	01/03/2016	Postal: P O BOX 5278, TYGER VALLEY, TYGER VALLEY, WESTERN CAPE, 7536 Residential: 33 SOMER STREET, EVERSDAL, EVERSDAL, WESTERN CAPE, 7550

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Physical Address
the dti Campus - Block F
77 Meintjes Street
Sunnyside 0001

Postal Address: Companies
P O Box 429
Pretoria
0001

Docex: 256
Web: www.cipc.co.za
Contact Centre: 086 100 2472 (CIPC)
Contact Centre (International): +27 12 394 9573

Attachment A1.2

SARS
South African Revenue Service

VALUE ADDED TAX
Notice of Registration

AFRIMAT CONTRACTING
PO BOX 505
WORCESTER
6849

Enquiries should be addressed to SARS

Contact Detail

SARS
Alberton
1528

0800 00 7277
Website: www.sars.gov.za

Details

Taxpayer Reference No: 4910205428

Always quote this reference number when contacting SARS

Date: 2023-10-23

Electronic Certification
SARS
South African Revenue Service
Certified as a true copy and there is no indication that the original has been amended by any unauthorised person
2023-10-23

Dear Taxpayer

NOTICE OF REGISTRATION

The South African Revenue Service (SARS) confirms registration of the following taxpayer:

Registered name: AFRIMAT CONTRACTING INTERNATIONAL (PTY) LTD
Trading name: AFRIMAT CONTRACTING INTERNATIONAL
Taxpayer registration number: 2002/025895/07
Taxpayer reference number: 4910205428
Date of Registration: 2003-04-01

Your tax obligation

The taxpayer has been registered for Value-Added Tax with effect from 20030401 and a 12 month(s) tax period with the following category has been allocated:

The last day of each of the twelve months of the calendar year.

The taxpayer must use the Invoice basis to account for the tax payable.

Payments in respect of VAT must be made under cover of a VAT 201 return to SARS by the 25th of the month after the end of the tax period. eFilers have until the last business day of the month to make payment. If payment is not paid in full by the due date, a ten percent penalty and interest at the prescribed rate is levied.

Where the payment date falls on a weekend or public holiday, the due date is the last preceding business day.

Kindly notify SARS of any change to your registered particulars within 21 business days of such change.

Should you have any queries please call the SARS Contact Centre on 0800 00 7277. Remember to have your taxpayer reference number at hand when you call to enable us to assist you promptly.

Sincerely

Attachment A1.3

DMR 37

DEPARTMENT: MINERAL RESOURCES
REPUBLIC OF SOUTH AFRICA

MINING RIGHT

Granted in terms of section 23(1) of the Mineral and Petroleum Resources Development Act, 2002
(Act No. 28 of 2002)

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Protocol No: 06 **/2014**
File Ref No LP30/5/1/2/2/10023MR
Application No

LET IT HEREBY BE MADE KNOWN:

THAT on this **12** day of **February** in the year **2014**, before me, **Mr Maboku Isaac Mangena** a Notary Public, duly sworn and admitted, residing and practising at **Polokwane**, in the **Limpopo** Province of South Africa, and in the presence of the subscribing competent witnesses, personally came and appeared:

Aaron Khathutshelo Kharivhe, Regional Manager, **Limpopo** Region of the Department of Mineral Resources, and as such in his / her capacity as the duly authorised representative of:

THE MINISTER OF MINERAL RESOURCES

The said Regional Manager, being duly authorised thereto under and by virtue of a Power of Attorney granted by the **Deputy Director-General** of the Department of Mineral Resources on the **11th** day of **December** in the year **2013** in terms of the powers delegated by the Minister on the 12th day of May 2004 in terms of section 103 (1) of the Act,

Mining Right: Granted in terms of Section 23 of the Mineral and Petroleum Resources Development Act, No. 28 of 2002

AND

Mr Johannes Mzamo Kalo in his/her personal capacity or as the company's Director or as a Close Corporations Member, and as such, the duly authorised representative of Afrimat Aggregate Trading (Pty) Ltd,Identification/ Registration number:

1	9	9	4	/	0	0	5	2	7	1	/	0	7	

(Hereinafter together with his/her/its successors in title and assigns referred to as "the Holder", he/she, the said representative, being duly authorised thereto under and by virtue of a power of attorney/resolution of directors/members of the Holder, signed or passed at Bellville on the 30 day of October in the year 2013 which power of attorney or a certified copy of a resolution has this day been exhibited to me, the notary, and remain filed of record in my protocol with the minutes hereof.)

AND THE MINISTER AND HOLDER DECLARED THAT:

WHEREAS The State is the custodian of the Nation's mineral and petroleum resources in terms of section 3 of the Act.

AND WHEREAS The Holder has applied for a mining right in terms of section 22 of the Act,

AND WHEREAS The **Deputy Director General** of the Department of Mineral Resources has by virtue of powers delegated to him, granted to the Holder, a mining right in terms of section 23(1) of the Act.

NOW THEREFORE THE MINISTER GRANTS A MINING RIGHT SUBJECT TO THE FOLLOWING TERMS AND CONDITIONS:

Definitions

In this mining right, the following words and expressions shall have the meanings assigned to them:

'***Act***' means the Mineral and Petroleum Resources Development Act, 2002 (Act 28 of 2002) and includes the Regulations, guidelines, circulars, directives and orders made in terms of that Act;

'***Effective date***' means **12th** day of **February** in the year **2014** (being the date on which the environmental management programme is approved in terms of section 39(4) of the Act);

'***Environmental Management Programme***' is as defined in the Act and includes any other Environmental Management Programme approved in terms of the previous mining legislation;

'***Financial year***' means a complete financial year of the Holder which, at the time of the granting of this mining right, commences on **01st** day of **March** in the year **2014**; and ends on **01st** day of **March** in the year **2014**;

'***Holder***' is as defined in the Act, and specifically in relation to this right, it means **Afrimat Aggregate Trading (Pty) Ltd**, Registration No/Identification No **199400527107**;

'***Mineral***' is as defined in the Act, and specifically in relation to this right means **Quartzitic Sandstone and Conglomerate**;

'***Mining Area***' is as defined in the Act and includes any additional area of environmental liability as may be reflected on the Environmental Management Programme relating to this right;

'***Mining right***' is as defined in the Act and includes all the Annexures to it, agreements and inclusions by reference;

'***Mining Work Programme***' is as defined in the Act and as reflected in the attached **Annexure A** to this mining right;

'***Minister***' means the Minister of Mineral Resources and includes the successors in title, the assignee or any person duly authorised to act in the Minister's place and stead;

'**Regional *Manager***' is as defined in the Act and specifically in relation to this right means the Regional Manager for the **Limpopo** Region of the Department of Mineral Resources; and

'***Social and Labour Plan***', is as contemplated in regulation 46 of the Regulations to the Act and is as reflected in the attached **Annexure B** to this mining right.

1. Description of the Mining Area

The Mining Area shall comprise the following:
Certain: **Remainder of the farm Kuipersbult 511 LQ**
Situated: **Lephalale,** Magisterial/Administrative District of **Lephalale**
Measuring: **25,4689** hectares in extent.
(In the case of various farms being involved, a list can be attached and referred to as **Annexure**);
Which Mining Area is described in detail on the attached Diagram/plan marked **Annexure** .

2. Granting of Mining Right

Without detracting from the provisions of sections 5 and 25 of the Act, the Minister grants to the Holder the sole and exclusive right to mine, and recover the mineral/s in, on and under the mining area for the Holder's own benefit and account, and to deal with, remove and sell or otherwise dispose of the mineral/s, subject to the terms and conditions of this mining right, the provisions of the Act and any other relevant law in force for the duration of this right.

3. Commencement, Duration and Renewal

3.1. This mining right shall commence on **12 February 2014** and, unless cancelled or suspended in terms of this clause 13 of this right and or section 47 of the Act, will continue to be in force for a period of **30** years ending on **11 February 2044**.

3.2. The Holder must commence with the mining operations within a year from the date on which the mining right becomes effective in terms of section 23 (5) of the Act, or any later date as may, upon a written request by the Holder, be authorised in writing by the Minister in terms of the Act, failing which this right may be cancelled or suspended.

3.3. Any application for renewal must be submitted to the Regional Manger not later than 60 working days prior to the date of expiry of this right.

4. Amendments, Variation and Abandonment

4.1. The terms of this right (including by extension of the area covered by it or by the addition of minerals or a share or shares or seams, mineralized bodies, or strata, which are not at the time the subject thereof) may not be amended or varied without the written consent of the Minister.

4.2. The Holder shall be entitled to abandon or relinquish the right or the area covered by the right entirely or in part. Upon abandonment or relinquishment of the mining area or any portion thereof, the Holder must:

4.2.1. Furnish the Regional Manager with all prospecting and /or mining results and/or information, as well as the general evaluation of the geological, geophysical and borehole data in respect of such abandoned area in so far as it applies to the mineral or any other mineral/s obtained in respect of this right and,

4.2.2. Apply for a closure certificate in terms of section 43 (3) of the Act.

4.3 With effect from the date the Holder has abandoned or relinquished a portion or portions of the mining area, and subject to section 43 of the Act, the Minister is entitled to grant any right, permit, or permission referred to in the Act in, on, or under the portion/s, so abandoned or relinquished, to any person/s.

5. Payment of Royalties

5.1. The Holder shall as contemplated in section 25 (2) (g) pay to the State throughout the duration of this mining right, royalties payable in terms of any Act or Amendment to an Act of Parliament implemented.

6. Payment of Interest

If mining fees, any fees, any levy, royalties or consideration referred to in clause 5 are not paid punctually, the Holder shall be in mora and shall pay interest thereon at the rate prescribed in terms of section 80 of the Public Finance Management Act, 1999(Act 1 of 1999) reckoned from the date on which payment is due and payable, to the date of actual payment.

7. Restrictions and Obligations Imposed on the Holder

7.1 The Holder is entitled to the rights referred to in section 5(2), (3) and section 25 of the Act, and such other rights as may be contained in this mining right or such other right as may be granted to, acquired by or conferred upon the Holder by any other applicable law.

7.2 Mining operations in the mining area must be conducted in accordance with the Mining Work Programme and any amendment to such Mining Work Programme and an approved Environmental Management Plan.

7.3 The Holder shall not trespass or enter into any homestead, house or its curtilage nor interfere with or prejudice the interests of the occupiers and/or owners of the surface of the Mining Area except to the extent to which such interference or prejudice is necessary for the purposes of enabling the Holder to properly exercise the Holder's rights under this mining right.

8. Conditions on disposal of Minerals and/ or Products Derived from Mining

It is a condition of the granting of this mining right that the Holder shall dispose of all minerals and/ or products derived from the exploitation of the mineral at competitive market prices which shall mean in all cases, non-discriminatory prices or non-export parity prices. If the minerals are sold to any entity, which is an affiliate or non-affiliated agent or subsidiary of the Holder, or is directly or indirectly controlled by the Holder, such purchaser must unconditionally undertake in writing to dispose of the minerals and any products produced from the minerals, at competitive market prices.

9. Mortgage, Cession, Transfer, Alienation

9.1 This mining right, a shareholding, an equity, an interest or participation in the right or joint venture, or a controlling interest in a company, close corporation or joint venture, may not be encumbered, ceded, transferred, mortgaged, let, sublet, assigned, alienated or otherwise disposed of without the written consent of the Minister, except in the case of a change of controlling interest in listed companies.

9.2 Any transfer, encumbrance, cession, letting, sub-letting, assignment, alienation or disposal of this right or any interest therein or any share or any interest in the Holder, without the consent of the Minister referred to in section 11(1) is of no force, no effect and is invalid.

10. Protection of Boreholes, Shafts, Edits and Openings.

All boreholes, shafts, edits, excavations, and openings sunk or made, by the Holder during the currency of this mining right shall be sealed, closed, fenced, made safe by the Holder in accordance with the approved Environmental Management Programme, the Mine Health and Safety Act, 1996 or any other applicable laws and Regulations.

11. Holder's liability for payment of Compensation for Loss or Damage

11.1. Subject to section 43 of the Act, the Holder shall, during the tenure of this right while carrying out the mining operations under this right, take all such necessary and reasonable steps to adequately safeguard and protect the environment, the mining area and any person/s using or entitled to use the surface of the mining area from any possible damage or injury associated with any activities on the mining area.

11.2. Should the holder fail to take reasonable steps referred to above, and to the extent that there is legal liability, the holder shall compensate such person or persons for any damage or losses, including but not limited to damage to the surface, to any crops or improvements, which such person or persons may suffer as a result of, arising from or in connection with the exercise of his/her rights under this mining right or of any act or omission in connection therewith.

12. Inspection of Mining Area

The Minister and/or any person duly authorised thereto in writing by the Minister shall be entitled to inspect the mining area, the Holder's mining operations and the execution of the approved Environmental Management Programme on the Mining Area as provided for in the Act, and any instruction conveyed in writing by the Minister to the Holder requiring the proper performance by the Holder of the Holder's obligations under this mining right shall be put into effect by the Holder in terms of the Act.

13. Cancellation or Suspension

13.1 Subject to section 47 of the Act, this mining right may be cancelled or suspended if the Holder:

13.1.1 Submits inaccurate, incorrect and or misleading information in connection with any matter required to be submitted under the Act;

13.1.2 Fails to honour or carry out any agreement, arrangement, or undertaking, including the undertaking made by the Holder in terms of the Broad Based Socio Economic Empowerment Charter and Social and Labour plan, on which the Minister relied for the granting of this right;

13.1.3 Breaches any material term and condition of this mining right;

13.1.4 Conducts mining operations in contravention of the provisions of the Act;

13.1.5 Contravenes the requirement of the approved Environmental Management Programme; or

13.1.6 Contravenes any provisions of this Act in any other manner.

13.2 Before the Minister cancels or suspends this right, the Minister shall:

13.2.I Give written notice to the Holder indicating the intention to suspend or cancel this right;

13.2.2 Give reason/s why the Minister is considering the suspension or cancellation of this right;

13.2.3 Give the Holder 30 days to show reasons why the right should not be suspended or cancelled;

13.2.4 Notify, the mortgagee [if any], of the intention to suspend or cancel this right; and

13.2.5 Direct the Holder, where it is possible to remedy any contravention, breach or failure, to comply or to take such specified measures to remedy any contravention, breach or failure to comply.

13.3 If the Holder does not take the measures as specified by the Minister to remedy a contravention, breach or failure, the Minister may cancel or suspend this right after considering representations made by the Holder in terms of clause 13.2.3.

BG

14. Records and Returns

14.1. The Holder shall maintain all such books, plans and records in regard to mining on the Mining Area as may be required by the Act and shall furnish to the office of the Regional Manager such reports and documents as may be relevant under this right.

14.2. The Holder shall furnish to the Regional Manager all such monthly returns contemplated in section 28 (2) A of the Act not later than the 15th day of the month following the month in respect of which it was reported.

14.3 The Holder shall furthermore at the end of each year following commencement of this mining right, inform the Regional Manager in writing of any new developments and of the future mining activities planned in connection with the exploitation/mining of the minerals on the Mining Area.

15. Minister's liability for Compensation

The Minister shall not at any time be liable or responsible for the payment of compensation of whatever nature to the Holder, the Holder's successors-in-title or assignee, or any person whomsoever as a result of the granting of this right.

16. Compliance with the Laws of the Republic

The granting of this Right, does not exempt the Holder and its successors in title and/or assigns from complying with the relevant provisions of the Mine Health and Safety Act, (Act No.29 of 1996) and any other law in force in the Republic of South Africa.

17. Provisions relating to section 2(d) and (f) of the Act

In the furthering of the objects of this Act, the Holder is bound by the provisions of an agreement or arrangement dated entered into between the Holder/ empowering partner and (the empowerment partner) which agreement or arrangement was taken into consideration for purposes of compliance with the requirements of the Act and or Broad Based Economic Empowerment Charter developed in terms of the Act and such agreement shall form part of this right.

18. Social and Labour Plan

18.1 The holder must annually, not later than three months before the end of its financial year, submit detailed implementation plan to give effect to Regulation 46(e)(i),(ii)and (iii) in line with the Social and Labour Plan.

18.2 The holder must annually, not later than three months after finalisation of its audited annual report, submit a detailed report on the implementation of the previous year's social and labour plan.

19. Severability

Notwithstanding anything to the contrary, any provision of this mining right which is contrary to any provision of the Act or which is otherwise ultra vires, null and void, voidable, or unenforceable, shall be severable from the rest of this right, such rest thus being and remaining of full force, effect and enforceable.

20. Domicilia citandi et executandi

20.1. The parties hereto choose the following addresses as their *domicilia citandi et executandi* and for all purposes arising from this mining right, in particular for the purposes of serving of any notice in terms of this mining right, and any notice properly addressed to the under mentioned postal addresses of the parties shall be deemed to have been received by the addressee within 14 days if given in writing and posted by prepaid registered post addressed to the addressee at the relevant postal address:

20.1.1. In the case of the ***Minister***:

Physical Address	Postal Address
101 Dorp Street **Polokwane** Code **0699** Tel **015 287 4700** Fax **015 287 4729**	**Private Bag 9467** **Polokwane** **0700** **015 287 4700** **015 287 47 29**

20.1.2. In the case of the ***Holder***:

Physical Address	Postal Address
Tyger Valley Office Park No 2 **Cnr Willie Van Schoor Avenue and Old Oak Road** **Bellville** Code **7535** Tel **021 9178 840** Fax **021 9178 822**	**P.O BOX 768** **Bellville** **7535** **021 9178 840** **021 9178 822**

20.2. Notwithstanding anything to the contrary herein contained, a written notice or communication actually received by a party at any place other than the chosen *domicilia citandi et executandi* shall constitute adequate notice or communication to the party notwithstanding that it was not sent to or delivered at such party's chosen *domicilium citandi et executandi.*

20.3 Either party shall be entitled from time to time to change the *domicilia citandi et executandi* or postal address furnished above after giving at least 14 days prior written notice of such change to the other party, failing which the above mentioned addresses will remain in force.

20.4. Any written notice or communication contemplated in this clause which is forwarded by one party to the other by registered post will be presumed to have been received by the addressee on the fourteenth day following the date of posting from an address within the Republic of South Africa to the addressee at the postal address of the addressee for the time being as determined in accordance with the provisions of this clause.

21. Costs

The Holder shall pay all costs and charges incurred in connection with the execution and registration of this prospecting right.

Thus done and signed at **Polokwane** on the **12** day of **February** in the year **2014** in the presence of the undersigned witnesses:

AS WITNESS:

For and on behalf of the **Minister**

AS WITNESS:

For and on behalf of the **Holder**

Notary Public

Attachment A1.4

water & sanitation

Department:
Water and Sanitation
REPUBLIC OF SOUTH AFRICA

Private Bag X313, Pretoria, 0001, Sedibeng Building, 185 Francis Baard Street, Pretoria,
Tel: (012) 336 7500, Fax (012) 323-4472 / (012) 326 - 2715

LICENCE IN TERMS OF CHAPTER 4 OF THE NATIONAL WATER ACT, 1998 (ACT NO. 36 OF 1998) (THE ACT)

I, ***Margaret-Ann Diedricks,*** in my capacity as Director-General in the Department of Water and Sanitation acting under authority of the powers delegated to me by the Minister of Water and Sanitation, hereby authorise the following water use in respect of this licence.

SIGNED: .....

DATE: 17 March 2016

LICENCE NO: 01/A42J/AGI/4378
FILE NO: 27/2/2/A942/4/1

1. **Licensee:** **AFRIMAT AGGREGATES (Trading) (Pty) Limited.**

   Postal Address: P.O. Box 5278
   Tyger Valley
   7536

2. **Water Use**

Section 21(a) of the Act:	Taking of water from a water resource, subject to the conditions set out in Appendices I and II
Section 21 (g) of the Act:	Disposing of waste in a manner which may detrimentally impact on a water resource, subject to the conditions set out in Appendices 1 and III
Section 21 (j) of the Act:	Removing, discharging or disposing of water found underground if it is necessary for the efficient continuation of an activity or for the safety of people; subject to the conditions set out in Appendices 1 and III

3. **Property in respect of which the license is issued**

   Afrimat aggregates (Trading) (Pty) limited.

B 07071

Page 1 of 12

LICENCE NO: 01/A42J/AGI/4378
FILE NO: 27/2/2/A942/4/1

4. Registered owner of the Property

Eskom Holdings Ltd

5. Licence and Review Period

This licence is valid for a period of twenty (20) years from the date of issuance and it may be reviewed on intervals of not more than five (5) years.

6. Definitions

Any terms, words and expressions as defined in the National Water Act, 1998 (Act 36 of 1998) shall bear the same meaning when used in this licence.

"The Provincial Head" means the Head of Provincial Operations: Limpopo, Department of Water and Sanitation, Private Bag X9506, Polokwane, 0700.

"Act" means National Water Act, 1998 (Act 36 of 1998)

"The Minister" means Minister of the Department of Water and Sanitation.

7. Description of the activity

This water use licence authorises an abstraction of 1 828 m^3/a (one thousand eight hundred and twenty eight cubic metres per annum) for the processing of quartz stones and for storing of pit water in the reservoir. It also relates to the stockpiling of fine material/silt generated from product processing on the 11.6 ha piece of land. The activity is on portion 0 of the farm Kuipersbult 511 LQ, Limpopo. The geographic location of the proposed activity is S 23° 43' 36.7" and E 27° 33' 29.3".

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LICENCE NO: 01/A42J/AGI/4378
FILE NO: 27/2/2/A942/4/1

APPENDIX I

General conditions for the licence

1. This licence is subject to all applicable provisions of the National Water Act, 1998 (Act 36 of 1998).
2. The responsibility for complying with the provisions of the licence is vested in the Licensee and not any other person or body.
3. The Licensee must immediately inform the Provincial Head of any change of name, address, premises and/or legal status.
4. If the property in respect of which this licence is issued is subdivided or consolidated, the Licensee must provide full details of all changes in respect of the properties to the Provincial Head of the Department within sixty (60) days of the said change taking place.
5. If a water user association is established in the area to manage the resource, membership of the Licensee to this association is compulsory.
6. The Licensee will be responsible for any water use charges or levies imposed by the Provincial Head.
7. While effect must be given to the Reserve as determined in terms of the Act, where a desktop determination of the Reserve has been used in issuance of a licence, when a comprehensive determination of the Reserve has finally been made; it shall be given effect to.
8. When compulsory licensing is implemented for the water resource in respect of which this licence was issued, the water use authorised in this licence could be subject to appropriate reduction.
9. The licence must not be construed as exempting the Licensee from compliance with the provisions of any other applicable Act, Ordinance, Regulation or By-law.
10. The licence and amendment of this licence are also subject to all the applicable procedural requirements and other applicable provisions of the Act, as amended from time to time.
11. The Licensee must conduct an annual internal audit on compliance with the conditions of licence. A report on the audit shall be submitted to the Provincial Head within one month of the finalisation of the audit.
12. The Licensee shall appoint an independent external auditor to conduct an annual audit on compliance with the conditions of this licence. The first audit must be conducted within 3 (three) months of the date this licence be issued and a report on the audit shall be submitted to the Provincial Head within one month of finalisation of the report.
13. Flow metering, recording and integrating devices must be maintained in a sound state of repair and calibrated by a competent person at intervals of not more than two years.

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LICENCE NO: 01/A42J/AGI/4378
FILE NO: 27/2/2/A942/4/1

Calibration certificates must be available for inspection by the Provincial Head or his/her representative upon request.

14. Any incident that causes or may cause water pollution must be reported to the Provincial Head or his/her designated representative within 24 hours.

Page 4 of 14 Afrimat Aggregates (Trading)(Pty) Ltd. Director-General

LICENCE NO: 01/A42J/AGI/4378
FILE NO: 27/2/2/A942/4/1

APPENDIX II

Section 21(a): Taking water from a water resource

1. This water use licence authorises an abstraction of 1 828 m^3/a (one thousand eight hundred and twenty eight cubic metres per annum) from a quarry pit for the processing of quartz stones at portion 0 of the farm kuipersbult 511 LQ, Limpopo. The geographic location of the proposed activity is S 23° 43' 36.7" and E 27° 33' 29.3".

2. The quantity of water authorised to be taken in terms of this licence may not be exceeded.

3. This licence does not imply any guarantee that the said quantities and qualities of water will be available at present or at any time in the future.

4. The above mentioned volume may be reduced when the licence is reviewed.

5. If the water use in this licence is not exercised within five (5) years from the date of issuance of this licence, the authorisation will be amended.

6. The Licensee must continually investigate new and emerging technologies and put into practice water efficient devices or apply technique for the re-use of water containing waste, in an endeavour to conserve water at all times.

7. The Licensee must install appropriate water measuring devices to measure the amount of water abstracted. The Licensee must ensure that all measuring devices are properly maintained and in good working order and must be easily accessible. This must include a programme of checking, calibration, and/ or renewal of measuring devices. All water taken from the resource must be measured; recorded and reported as follows:

   7.1. The daily quantity of water taken must be metered or gauged and the total recorded at the last day of each month; and

   7.2. The Licensee must keep record of all water measurements taken and a copy of the records must be forwarded to the Provincial Head on or before 25 January and 25 July of each year.

8. No water taken may be used for purposes other than intended in this licence, without written approval by the Provincial Head or his/her delegated nominee.

9. Notices prohibiting unauthorised persons from entering the certain areas, as well as internationally acceptable signs indicating the risks involved in case of an unauthorised entry must be displayed along the boundary fence of these areas.

Page 5 of 14 Afrimat Aggregates (Trading)(Pty) Ltd. Director-General

LICENCE NO: 01/A42J/AGI/4378
FILE NO: 27/2/2/A942/4/1

10. The Department accepts no liability for any damage, loss or inconvenience, of whatever nature, suffered as a result of:
    10.1. shortage of water;
    10.2. inundations or flood;
    10.3. siltation of the resource; and
    10.4. required reserve releases.

11. The Licensee must establish a programme of formal Information Management System, which maintains a database on water supply, distribution and delivery infrastructure.

12. The Licensee must establish and implement a continual process of raising awareness amongst itself, its workers and stakeholders with respect to Water Conservation and Water Demand Management initiatives.

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APPENDIX III

Section 21(g) of the Act: Disposing of waste in a manner which may detrimentally impact on water resource

1. CONSTRUCTION, OPERATION AND MAINTANANCE

1.1 The licensee must carry out and complete all the activities, including the construction of disposal facilities as listed in Table 2 below according to the Report and according to the final plan as described in the Report

1.2 The waste facilities must be operated and maintained to have a minimum freeboard of 0.8 metres above full supply level and all other water systems related thereto must be operated in such a manner that it is at all times capable of handling the 1:50 year flood-event on top of its mean operating level.

1.3 The construction of Dirty water disposal facilities must be carried out under the supervision of a professional Civil Engineer registered under the Engineer Profession of South Africa Act, 1990 (Act 114 of 1990, as approved by the designer.

1.4 Within 30 days after the completion of the activities referred here in accordance with the relevant provisions of this licence, the licensee must in writing; under reference 27/2/2/A942/4/1 informs the Provincial Head thereof. This must be accompanied by a signature of approval from the designer referred to above that the construction was done according to the plans referred in the Report.

1.5 The licensee must ensure that the disposal of the polluted water and the operation and maintenance of the system are done according to the provisions in the IWULA and the permit issued by the Department of Mineral Resources.

2. DISPOSAL OF WATER CONTAINING WASTE

2.1 The Licensee is authorised to dispose waste or water containing waste into the waste management facilities on the property as indicated in Table 2.

Table 2: Volumes of waste to be disposed at the waste disposal facilities

Facility	Quantity	Co-ordinate	Property Name
Storage Dam containing pit water	53 000 m^3/a	S:23°43'36.7" E:27°33'29.3"	Portion 0 of the farm Kuipersbult 511

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3 GROUNDWATER MONITORING

3.1 The applicant must conduct and complete a groundwater investigation study to determine the impact on ground water and any mitigation actions that could be required. This must be submitted to the Provincial Head within one (1) year from the date of the issuance of this licence.

3.2 The Licensee must develop a groundwater monitoring network and which must be set as an early warning system to detect any pollution caused by seepage from the irrigation field.

3.3 The groundwater quality monitoring must conducted on a quarterly basis on boreholes in groundwater monitoring network referred to in condition 2.2 in Appendix III. The water samples should be analysed for the parameters indicated in Table 1 below and any other parameters of concern.

Table 3: Variables to be monitored and Monitoring Frequency

Parameter	Frequency
pH	Quarterly
Electrical Conductivity(mS/m)	Quarterly
Sodium (mg/l)	Quarterly
Magnesium (mg/l)	Quarterly
Sulphate (mg/l)	Quarterly
Faecal coliform (count/100ml)	Quarterly
Chloride (mg/l)	Quarterly
Nitrate (mg/l)	Quarterly
Fluoride (mg/l)	Quarterly

3.4 If ground water pollution have occurred or may possibly occur the Licensee must conduct necessary investigation and implement additional monitoring and rehabilitation measures which must be the satisfaction of the Provincial Head

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4. WATER RESOURCE PROTECTION

4.1 The impact of the activities of the industry on the groundwater must not exceed the groundwater quality limits as indicated in Table 4

Table 4: Water Quality Reserve

Parameter	Water Quality Reserve
Electrical Conductivity (mS/m)	59.4
Sodium (mg/l)	31.0
Magnesium (mg/l)	16.0
Calcium (mg/l)	31.0
Chloride (mg/l)	28.0
Sulphate (mg/l)	10.3
Nitrate(mg/l)	0.8
Fluoride(mg/l)	0.4
pH	7.8

4.2 The Licensee must come up with alternative effective wastewater treatment technologies that are environmental friendly (i.e. package plant or conservancy tanks) and decommission the french drain systems within 12 months from the date of issuance of this licence.

5. INCIDENT MONITORING

5.1 If surface and/or groundwater pollution has occurred or may possibly occur, the License must conduct, and/or appoint specialists to conduct the necessary investigations and implement additional monitoring, pollution prevention and remediation measures to the satisfaction of the Provincial Head.

5.2 The Licensee must keep all records relating to the compliance or non-compliance with the conditions of this licence in good order. Such records must be made available to the Provincial Head within fourteen (14) days of receipt of a written request by the Department for such records.

5.3 The Licensee must keep an incident report and complaints register, which must be made available to any external auditors and the Department.

5.4 Pollution incidents must be dealt with in accordance with the Act.

6. REPORTING

6.1 The Licensee must update the water balance annually and calculate the loads of waste emanating from the activities. The Licensee must determine the contribution of their activities to the mass balance for the water resource and must furthermore co-operate with other water users in the catchment to determine the mass balance for the water resource reserve compliance point.

6.2 The Licensee must submit the results of analysis for the monitoring requirements to the Provincial Head on a quarterly basis under Reference number **27/2/2/A942/4/1**

7. BUDGETARY PROVISIONS

7.1 The Licensee must ensure that there is a budget sufficient to complete and maintain the water use and for successful implementation of the rehabilitation programme as set out in this licence.

7.2 The Provincial Head may at any stage of the process request proof of budgetary provisions for rehabilitation and closure of project.

8. STORMWATER MANAGEMENT

8.1 Stormwater leaving the Licensee's premises shall in no way be contaminated by any substance, whether such substance is a solid, liquid, vapour or gas or a combination thereof which is produced, used, stored, dumped or spilled on the premises.

8.2 Increase runoff due to vegetation clearance and/or soil compaction must be managed, and steps must be taken to ensure that stormwater does not lead to bank instability and excessive levels of silt entering the stream.

8.3 Stormwater shall be diverted from the industrial complex site and roads and shall be managed in such a manner as to disperse runoff and concentrating the stormwater flow.

8.4 Where necessary works must be constructed to attenuate the velocity of any stormwater discharge and to protect the banks of the affected watercourses.

8.5 Stormwater control works must be constructed, operated and maintained in a sustainable manner throughout the impacted area.

8.6 All stormwater that would naturally run across the pollution areas shall be diverted via channels and trapezoidal drains designed to contain the 1:50 year flood.

8.7 Polluted stormwater captured in the stormwater control dams must be pumped to the process water treatment plant for reuse and recycling.

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9 PLANT AREAS AND CONVEYANCES

9.1 Pollution caused by spills from the conveyances must be prevented through proper maintenance and effective protective measures especially near all stream crossings.

9.2 All reagent storage tanks and reaction units must be supplied with a bunded area built to the capacity of the facility and provided with sumps and pumps to return the spilled material back into the system. The system must be maintained in a state of good repair and standby pumps must be provided.

9.3 Any hazardous substances must be handled according to the relevant legislation relating to the transport, storage and use of the substance.

9.4 Any access roads or temporary crossings must be:

9.4.1 non-erosive, structurally stable and must not induce any flooding or safety hazard and be repaired immediately to prevent further damage.

10 ACCESS CONTROL

10.1 Strict access procedures must be followed in order to gain access to the property. Access to waste or water containing waste facilities must be limited to authorised employees of the Licensee and their Contractors only.

10.2 Notices prohibiting unauthorised persons from entering the controlled access areas as well as internationally acceptable signs indicating the risks involved in case of an unauthorised entry must be displayed along the boundary fence of these areas.

11 CONTINGENCIES

11.1 Accurate and up-to-date records must be kept of all system malfunctions resulting in non-compliance with the requirements of this licence. The records must be available for inspection by the Provincial Head upon request. Such malfunctions must be tabulated under the following headings with a full explanation of all the contributory circumstances:

11.1.1 operating errors
11.1.2 mechanical failures (including design, installation or maintenance)
11.1.3 environmental factors (e.g. flood)
11.1.4 loss of supply services (e.g. power failure) and
11.1.5 Other causes.

12 REPORTING OF EMERGENCIES

12.1 The Licensee must, within 24 hours, notify the Provincial Head of the occurrence or potential occurrence of any incident which has the potential to cause, or has caused

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water pollution, pollution of the environment, health risks or which is a contravention of the licence conditions.

12.2. The Licensee must, within 14 days, or a shorter period of time, as specified by the Provincial Head, from the occurrence or detection of any incident referred above, submit an action plan, which must include a detailed time schedule, to the satisfaction of the Provincial Head of measures taken to:

12.3.1 Correct the impacts resulting from the incident
12.3.2 Prevent the incident from causing any further impacts and
12.3.3 Prevent a recurrence of a similar incident.

13. AUDITING

13.1 The licensee must appoint an independent external auditor to conduct an annual audit on compliance with the conditions of this licence. The first audit must conducted within 3 (three) months of the date this licence was issued and a report on the audit shall be submitted to the Provincial Head within one month of finalisation of the report.

14. INTEGRATED WATER AND WASTE MANAGEMENT

14.1 The Licensee must update an *Integrated Water and Waste Management Plan (IWWMP),* which must together with the updated *Rehabilitation Strategy* and *Implementation Programme (RSIP),* be submitted to the Provincial Head for approval within one (1) year from the date of issuance of this licence.

14.2 The IWWMP and RSIP shall thereafter be updated and submitted to the Provincial Head for approval, annually.

14.3 The Licensee must, at least 180 days prior to the intended closure of any facility, or any portion thereof, notify the Provincial Head of such intention and submit any final amendments to the IWWMP and RSIP as well as a final *Closure Plan,* for approval.

14.4 The Licensee must make full financial provision for all investigations, designs, construction, operation and maintenance for a water treatment plant should it become a requirement as a long-term water management strategy.

15 SITE SPECIFIC CONDITIONS

15.1 The Licensee must submit the mining rights to the Department within twelve (12) months from the date of issuance of this water use licence.

15.2 The Licensee must identify all possible source of pollution and the engineering mitigation measures formulated and submit to the Department. Methods to stop migration of identified pollution should be implemented to ensure that pollution is kept within the boundaries of the activity.

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APPENDIX IV

Section 21(j) of the Act: Removing, discharging or disposing of water found underground if it is necessary for the efficient continuation of an activity or for the safety of people.

1 REMOVAL OF WATER FOUND UNDERGROUND

1.1 The Licensee is authorized to remove or dewater a maximum quantity of water in the properties and with a purpose as indicated in Table 5.

Table 5: Removal of water found underground

Description of Activity / Purpose	Purpose	Property	Volume (m^3/a)	Coordinates
Removal of water from open pit at the Kuipersbult quarry.	Plant processing, dust suppression and ablution facility	Portion 0 of the farm Kuipersbult 511	35 000	S:23°43'36.7" E:27°33'29.3"

1.2. The quantity of the water authorised to be removed and disposed of into the storage dam and process plant in terms of this licence may not be exceeded without prior authorisation by the Minister.

1.3. The quantity of water removed from underground must be metered and recorded on a daily basis.

1.4. The groundwater levels must be monitored every six (6) months (once in the beginning of the dry season and once in the beginning of the wet season).

1.5. Self-registering flow meters must be installed in the delivery lines at easily accessible positions near the dewatering points.

1.6. Calibration certificates in respect of the pumps must be submitted to the Provincial Head after installation thereof and thereafter at intervals of two years.

1.7. The date and time of monitoring in respect of each sample taken shall be recorded together with the results of the analysis.

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1.8. Analysis shall be carried out in accordance with methods prescribed by and obtainable from the South African Bureau of Standards, in terms of the Standards Act, 1982 (Act 30 of 1982).

1.9. The methods of analysis shall not be changed without prior notification to the Licensee and written approval by the Minister or his/her delegated nominee.

1.10. The Provincial Head must be informed of any incident that may lead to groundwater being disposed of contrary to the provisions of this licence, by submitting a report containing the following information:

1.10.1. Nature of the incident (e.g. operating malfunctions, mechanical failures, environmental factors, loss of supply services, etc);
1.10.2. Actions taken to rectify the situation and to prevent pollution or any other damage to the environment; and
1.10.3. Measures to be taken to prevent re-occurrence of any similar incident.

1.11. The Licensee must follow acceptable construction, maintenance and operational practices to ensure the consistent, effective and safe performance of the groundwater removal system.

1.12. Reasonable measures must be taken to provide for mechanical, electrical or operational failures and malfunctions of the underground water removal system.

[END OF LICENCE]

Attachment A1.5

Attachment A1.6